
A Greedy PDE Router for Blending Neural Operators and Classical Methods

Anonymous Author(s)

Affiliation

Address

email

1 **Revision note.** All text, tables and figures added in this revision are typeset in red (new figures
2 carry a red frame); everything in black is unchanged from the submitted version. The iteration-based
3 experiments of the submitted version on 31^2 and 63^2 grids with the LSTM router, and their appendix
4 analyses, have been removed: they are superseded by the wall-clock study of Sections E and 7,
5 which covers the same solvers and equations (plus anisotropic and variable-coefficient diffusion) on
6 128^2 – 512^2 grids and also reports the iteration-based metrics. Only the Dirichlet experiment of the
7 submitted version is kept (Section D).

8 **Status of the confirmatory re-run (draft for review).** In response to three rounds of review, every
9 wall-clock result is being regenerated under a revised, frozen protocol (Section E.1): compiled
10 classical kernels, a corrected forcing sampler, per-operation costs measured in the form in which
11 they are charged, the deployed rule equal to Algorithm 1 on macro-actions, HINTS with its two-
12 dimensional period $\tau = 15$, residual-decay controls, classical baselines timed in the same session
13 as the policies, a drift guard, two-sided tests with bootstrap intervals, a fresh test seed and three
14 warmed-up timed replays. At the time of this draft the following are *final*: the theory corrections
15 (Sections 4 and 5 and their proofs) and the wording of every claim, which is macro-driven and will
16 be recomputed from the complete tables. Everything marked **[pending]** in the text or shown as a
17 **[results pending]** table is still running, in this order: the 128^2 pairwise cells with their baselines;
18 the 128^2 ensembles; all 256^2 and 512^2 cells; usage traces, seed trials, overheads, the assumption,
19 screening and theorem checks and the discretisation study; variable-coefficient diffusion at 128^2 and
20 256^2 (Section E.5); and the decision-granularity ablation (Section E.6). Summary ranges quoted
21 in the abstract and in Section 7 are computed from the cells available at build time. Where a first
22 confirmatory run (before the protocol revision) had produced a number that is quoted in the text, it is
23 labelled as such.

Abstract

24 When solving PDEs, classical numerical solvers are often computationally ex-
25 pensive, while machine learning methods can suffer from spectral bias, failing to
26 capture high-frequency components. Designing an optimal hybrid iterative solver–
27 where, at each iteration, a solver is selected from an ensemble of solvers to leverage
28 their complementary strengths–poses a challenging combinatorial problem. While
29 greedy selection is desirable for its constant-factor approximation guarantee to the
30 optimal solution under Lipschitz and **shared-eigenbasis** assumptions, it requires
31 knowledge of the true error at each step, which is unavailable in practice. We
32 address this by proposing an approximate greedy router that efficiently mimics
33 a greedy approach to solver selection. Empirical results on the Poisson, convec-
34 tion–diffusion and anisotropic diffusion equations show that our method reduces
35 the area-under-the-curve (AUC) of the error trajectory relative to single-solver
36 baselines and existing hybrid approaches such as HINTS in every setting, reaching
37 comparable error levels in substantially fewer iterations with more stable error
38 decay. **With a cost-aware instantiation of the greedy rule, a lightweight router, and**
39 **a band-limited scale-equivariant corrector, we also measure end-to-end wall-clock**

time on 128^2 to 512^2 grids with compiled classical kernels, four equations (Poisson, convection–diffusion, anisotropic and variable-coefficient diffusion) and classical solvers from Jacobi to geometric multigrid, against HINTS, the best fixed schedule and the best residual-decay rule chosen per setting on the test set, and multigrid, preconditioned Krylov and direct solves timed in the same session. At truncation-level accuracy the learned router reproduces, without any search, the best static schedule found by a test-set search (one corrector call, then smoothing), to within [pending]–[pending] of it, and is [pending]–[pending] faster than HINTS with its published period; below truncation level it tracks the cost-aware oracle and beats HINTS and the one-shot schedule by large factors, but it is faster than the best fixed schedule in only [pending] of [pending] pairings at 128^2 , and a compiled geometric multigrid remains the strongest classical method there; routing over ensembles of classical solvers is free at truncation level and, at the coarse decision granularity that cheap compiled sweeps impose, gains nothing over the best single solver at 128^2 [256², 512² and finer-granularity cells pending].

1 Introduction

Natural phenomena and engineered systems are often governed by ordinary and partial differential equations (PDEs). Solving these equations enables a variety of tasks - predicting the evolution of the system through simulation (e.g., forecasting weather using the Navier-Stokes equations) [Kalnay, 2003, Bauer et al., 2015, Staniforth and Côté, 1991], addressing control problems (e.g., optimizing heat shield design for spacecrafts using heat transfer equations) [Anderson, 1989, Tröltzsch, 2010], and tackling inverse problems based on external measurements (e.g., reconstructing brain activity from EEG data using electrophysiological models) [Baillet et al., 2001, Grech et al., 2008].

Traditionally, PDEs are solved using finite difference methods [Smith, 1985, LeVeque, 2007], which discretize the spatial and/or temporal domain and approximate the solution by solving a system of equations at the discrete grid points. Similarly, finite element methods [Hughes, 2003, Bathe, 2006, Brenner, 2008] construct a system of equations based on local basis functions, and then rely on numerical linear algebra techniques—such as the Jacobi and Gauss-Seidel method [Saad, 2003]—to compute the solution. These iterative methods, however, **must be re-run from scratch for every new** initial condition, boundary condition, or forcing function, as even minor changes to any of these parameters require re-solving the entire system of equations from scratch.

These challenges have motivated the development of neural operators [Kovachki et al., 2023]—a class of machine learning models that aim to learn the solution operator directly, enabling fast inference across a range of parameters. Neural operators lift the classical linear layer in neural networks to an operator—typically a kernel integral operator—between function spaces. Despite strong approximation guarantees [Chen and Chen, 1995], neural operators are prone to spectral bias [Liu and Cai, 2021, Liu et al., 2024, You et al., 2024, Khodakarami et al., 2025, Xu et al., 2025], similar to traditional neural networks [Rahaman et al., 2019, Xu et al., 2019, Luo et al., 2019]. As a result, they favor learning low-frequency components of the solution function and often struggle to capture high-frequency features that iterative solvers excel at.

Recognizing this complementarity, Zhang et al. [2024] proposed HINTS, a hybrid solver that interleaves a neural operator pass every τ^{th} iteration of a classical solver, with τ fixed a priori. While HINTS reports significant empirical gains in convergence and accuracy over the standalone neural operator and Jacobi solver, this fixed schedule can be detrimental: a poorly timed neural correction may increase the error and undo recent progress.

Although state-dependent solver schedules are desirable, this naturally suggest a reinforcement learning formulation for learning routing policies. However, in our setting, the structure of iterative PDE solvers allows for a simpler approach: each solver induces a known update, and its effect can be evaluated through error-based signals. Leveraging this, we adopt a greedy rule that selects, at each iteration, the solver with the largest immediate error reduction. Such a rule can be near-optimal when the final error satisfies (weak) supermodularity—so that applying a beneficial solver earlier is at least as valuable as applying it later. To support this approach, we make the following contributions:

- We present a general hybrid PDE solver in which a routing rule chooses, at each iteration, from a set of classical solvers and neural operators. With oracle access to the true error at

every step, we show that a greedy routing rule achieves a constant-factor approximation to the optimal strategy for linear PDEs, provided the updates are error-reducing (Lipschitz with constant < 1) and zero-preserving—conditions under which weak supermodularity holds—and share a unitary eigenbasis, which gives the postfix monotonicity the guarantee also needs; for other ensembles the guarantee carries a measured slack.

- Given that the true error is not observed at test time, a convex surrogate loss is introduced, which, when minimized, enables the learned model to imitate the greedy router without access to true error information. This alignment is guaranteed through Bayes consistency.
- We empirically demonstrate that our approximate greedy solver achieves faster convergence than HINTS and individual solvers on the Poisson, convection-diffusion, anisotropic and variable-coefficient diffusion equations, in iterations and in end-to-end wall-clock time on 128^2 to 512^2 grids with compiled classical kernels and with the classical baselines timed in the same session. At truncation-level accuracy the router reproduces the best static schedule (one corrector call, then smoothing) without a search and is [pending]–[pending] faster than HINTS with its published period; below truncation level it beats HINTS and the one-shot schedule by large factors, whereas against the best fixed schedule and a residual-decay rule, both selected on the test set, the outcome depends on the pairing ([pending] of [pending] pairings at 128^2), and geometric multigrid alone remains faster there. The construction extends to ensembles of classical solvers, where we find no gain at the decision granularity the compiled kernels impose.

2 Background

Consider the following linear PDE:

$$\mathcal{L}_x^a(u) = f, x \in D; \quad \mathcal{B}_x(u) = b, x \in \partial D \quad (1)$$

where $\mathcal{L}_x^a$ is a differential operator with respect to the spatial variable $x \in D$ parameterized by the coefficient function a , f is a forcing function, u is the PDE solution, $\mathcal{B}_x$ is the boundary operator, and b is the boundary term. Such PDEs can be solved numerically using various discretization strategies, such as finite differences [Smith, 1985, LeVeque, 2007], finite element, and spectral methods [Boyd, 2001, Canuto et al., 2006]. Here, we focus on finite differences, which replace derivatives with difference quotients (e.g., $\partial_x u(x) \approx (u(x+h) - u(x))/h$) on a uniform grid.

Formally, let h be the grid size and $G_h(D)$ be a uniform grid with spacing h over D . For a function $g : D \rightarrow \mathbb{R}$, we denote its restriction to $G_h(D)$ by $g_h \in \mathbb{R}^N$, where $N = |G_h(D)|$. If $\mathcal{L}_h^a \in \mathbb{R}^{N \times N}$ is the discretized operator, the PDE reduces to a linear system of equations

$$\mathcal{L}_h^a u_h = f_h, \quad (2)$$

with the boundary conditions incorporated in $\mathcal{L}_h^a$. Unlike much of the neural operator literature, which retains function space formulations for discretization invariance, we follow the conventions of classical numerical analysis and represent functions and operators as vectors and matrices. Discretization invariance is not essential here, since we focus on fixed-grid problems.

2.1 Iterative PDE Solvers

Direct methods like Gaussian elimination and the Thomas algorithm can be computationally expensive in high-dimensional domains when solving systems like Equation (2). In contrast, iterative methods like Jacobi and Gauss-Seidel offer computational speedups by iteratively updating the solution, gradually converging to the true solution for the PDE. The general iterative update is

$$u^{(t+1)} = u^{(t)} + C \left(f_h - \mathcal{L}_h^a u^{(t)} \right), \quad (3)$$

where $u^{(t)}$ is the t^{th} iterate of the solution and C is a preconditioning matrix. Jacobi uses $C = D^{-1}$, where D is the diagonal of $\mathcal{L}_h^a$, and Gauss-Seidel uses $C = (D + L)^{-1}$, where L denotes the strict lower triangular part. However, these methods tend to damp low frequency parts of the error slowly.

2.2 Neural Operators

Suppose we observe samples $\{(a_i, f_i, u_i)\}_{i=1}^N$ such that $(a_i, f_i) \sim \mathcal{P}$ are i.i.d. samples, and $u_i = \mathcal{G}^*(a_i, f_i)$ be generated by a deterministic solution operator $\mathcal{G}^*$. A neural operator (NO) $\mathcal{G}_\theta$ seeks to approximate $\mathcal{G}^*$ by minimizing the expected squared $L^2(D)$ error:

$$\mathcal{R}_{\text{NO}}(\mathcal{G}_\theta) = \mathbb{E}_{(a,f) \sim \mathcal{P}} \left[\int_D \|\mathcal{G}^*(a, f)(x) - \mathcal{G}_\theta(a, f)(x)\|_2^2 dx \right]$$

Neural operators use discretization-invariant layers. Prominent instantiations include Fourier Neural Operators (FNO), which use spectral transforms [Li et al., 2020a], Graph Neural Operators, which perform graph-based aggregation over sampled points [Li et al., 2020b], and DeepONets, which employ a trunk-branch decomposition to map input functions to output functions [Lu et al., 2021].

2.3 Greedy Optimization

Consider maximizing $g(S)$ over $S \subseteq \Omega$ with $|S| \leq T$ where $g : 2^\Omega \rightarrow \mathbb{R}$ is a set function defined on subsets of a set Ω . An exhaustive search is infeasible, so an efficient alternative is the greedy algorithm. It builds the solution subset iteratively by adding ω^* that yields the largest marginal gain, i.e. $\omega^* = \arg \max_{\omega \in \Omega \setminus S} g(S \cup \omega)$. When g is non-negative, monotone (adding elements never decreases the value), and submodular (diminishing returns), the greedy algorithm achieves a $(1 - e^{-1})$ approximation to the optimal solution [Nemhauser et al., 1978]. Formally, a function g is submodular if $g(A \cup \{\omega\}) - g(A) \geq g(B \cup \{\omega\}) - g(B)$ for all $A \subseteq B \subseteq \Omega$ and $\omega \in \Omega \setminus B$; intuitively, delaying an addition cannot increase its benefit. For set function minimization problems, supermodularity, where the inequality flips, plays an analogous role, and the greedy rule achieves constant-factor approximation guarantees [Liberty and Sviridenko, 2017].

The suboptimality of the greedy algorithm has been extensively studied for both set-function minimization [Bounia and Koriche, 2023] and maximization [Das and Kempe, 2018, Feige et al., 2011, Bian et al., 2017, Harshaw et al., 2019] when standard assumptions—non-negativity, monotonicity, and sub/supermodularity—are weakened. In contrast, results on greedy sequence maximization [Streeter and Golovin, 2008, Alaei et al., 2021, Zhang et al., 2015, Bernardini et al., 2020, Van Over et al., 2024, Tschitschek et al., 2017]—where the ordering of the elements affects the function—have led to sequential analogues of submodularity and monotonicity. We leverage these tools to analyze the suboptimality of the greedy solution to minimizing the final error.

3 General framework for Hybrid Solvers

To solve Equation (2), consider the following hybrid iterative update:

$$u_h^{(t+1)} = u_h^{(t)} + C_{S_t} \left(f_h - \mathcal{L}_h^a u_h^{(t)} \right) \quad (4)$$

where $\mathcal{C} = \{C_j\}_{j=1}^K$ is a set of preconditioning functions and $S_t \in [K] = \{1, \dots, K\}$ indexes the function chosen at step t . Here, we use “preconditioning function” broadly to include classical linear updates ($C_j(x) = C_j x$ where C_j is a matrix), and learned models, such as neural operators. This update generalizes the classical iterative update in Equation (3), by allowing C_j to be non-linear and enabling the preconditioning function to be adaptively chosen at every step. $\mathcal{C}$ can also accommodate parameterized solver families (e.g., different Jacobi relaxation weights), enabling adaptive selection of solver parameters. As the number of solvers K grows, the chance of selecting a more effective update increases. Furthermore, HINTS [Zhang et al., 2024] is a special case of this hybrid solver with $K = 2$, where C_1 is a neural operator and C_2 is a classical preconditioner. Its routing rule is given by $S_t = \mathbf{1}_{(t+1) \bmod \tau > 0} + 1$, which selects C_1 every τ steps (first at iteration τ , after $\tau - 1$ classical steps, as in Algorithm S3 of Zhang et al. [2024]) and C_2 otherwise.

Let $e_h^{(t)} = u_h - u_h^{(t)}$ denote the error at step t . Then,

$$e_h^{(t+1)} = u_h - u_h^{(t)} - C_{S_t} \left(\mathcal{L}_h^a u_h - \mathcal{L}_h^a u_h^{(t)} \right) = (I - C_{S_t} \circ \mathcal{L}_h^a) (e_h^{(t)})$$

where I is the identity map and $I - C_j \circ \mathcal{L}_h^a$ is the error propagation function for solver j . Here, “ $\circ$ ” denotes function composition (matrix multiplication for linear updates). The objective is to select a

sequence $S = (S_1, \dots, S_T)$ that minimizes the error norm after T steps or $\|e_h^{(T)}\|_2^2$:

$$\min_{S_t \in [K], |S| \leq T} h(S) := \left\| (I - C_{S_{|S|}} \circ \mathcal{L}_h^a) \circ \dots \circ (I - C_{S_1} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \quad (5)$$

with compositions applied from right to left, so that the S_1 update acts on the initial error.

4 Greedy Algorithm

Equation (5) defines a combinatorial optimization problem with a search space exponential in T , making exact optimization intractable for large T or K . As a starting point, we consider an “omniscient” greedy algorithm that assumes access to the true initial error $e_h^{(0)}$. This is unrealistic since knowledge of $e_h^{(0)}$ could directly recover the solution via $u_h = u_h^{(0)} + e_h^{(0)}$, but it provides a useful benchmark. In Section 5, we relax this assumption using a practical learning strategy that is Bayes consistent with this omniscient rule: **in the infinite-sample limit it recovers the greedy decision at every state it is trained on, which is a per-decision statement; the sequential guarantee below is stated for the rule with access to the true error.**

As discussed in Section 2.3, when supermodularity and monotonicity hold, the greedy rule—such as the one described in Algorithm 1—enjoys constant-factor approximation guarantees. However, classical results focus on *set* functions, with only recent extensions made to sequences. Building on this line, we introduce a sequence-based notion of weak supermodularity.

Algorithm 1 Greedy Algorithm for a Hybrid PDE solver

Require: $\{C_j\}_{j=1}^K, T, \mathcal{L}_h^a, e_h^{(0)}$
 $S^0 \leftarrow \emptyset$
for $t < T$ **do**
 $S^{t+1} \leftarrow S^t \oplus \arg \min_{j \in [K]} \|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$
 $e_h^{(t+1)} \leftarrow (I - C_{S_{t+1}} \circ \mathcal{L}_h^a)(e_h^{(t)})$
end for
return S^T

Let Ω^* denote the space of sequences with elements in Ω . For $S, S' \in \Omega^*$, we denote their concatenation as $S \oplus S'$. S is a prefix of S' or $S \preceq S'$ if $S' = S \oplus L$ for some $L \in \Omega^*$. A function $g : \Omega^* \rightarrow \mathbb{R}$ is prefix monotonically non-increasing if $g(S \oplus S') \leq g(S)$ for all $S, S' \in \Omega^*$, and postfix monotonically non-increasing if $g(S' \oplus S) \leq g(S)$ for all $S, S' \in \Omega^*$. A prefix non-increasing function g is sequence supermodular if, for all $S', S \in \Omega^* : S \preceq S'$, it holds that

$$g(S) - g(S \oplus \omega) \geq g(S') - g(S' \oplus \omega), \quad \forall \omega \in \Omega \quad (6)$$

However, $h(S)$ (defined in Equation (5)) may not satisfy this property in general. Therefore, we introduce weak sequence supermodularity. A prefix non-increasing function g is weakly supermodular with respect to $S' \in \Omega^*$ if, for any $S \in \Omega^*$, there exists $\alpha(S') \geq 1$ such that

$$g(S) - g(S \oplus S') \leq \alpha(S') \sum_{i \in [|S'|]} g(S) - g(S \oplus S'_i) \quad (7)$$

The parameter $\alpha(S')$ or the supermodularity ratio quantifies deviation from exact sequence supermodularity. Expanding the $g(S) - g(S \oplus S')$ as a telescoping sum $\sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) - g(S \oplus (S'_1, \dots, S'_i))$ shows that the marginal decrease from appending S'_i after its predecessors is controlled by the effect of appending S'_i directly to S . Thus, postponing the inclusion of S'_i cannot yield a significantly larger benefit than adding it earlier. The supermodularity ratio $\alpha(S)$ quantifies the extent to which future gains from delays may exceed immediate gains.

Postfix monotonicity enters the analysis only through the greedy prefixes, so we also allow a slack: g is μ -postfix monotone with respect to O along a sequence S^T if $g(S^t \oplus O) \leq \mu g(O)$ for every prefix $S^t, t < T$; a postfix non-increasing function satisfies this with $\mu = 1$, and μ can be computed exactly for short horizons (Section E.10). Having introduced these notions, we now characterize the suboptimality of greedy solutions of weakly supermodular and postfix monotonic sequence functions in Theorem 4.1.

Theorem 4.1. Let $g : \Omega^* \rightarrow \mathbb{R}$ be a non-negative, prefix non-increasing function that is weakly supermodular with respect to the optimal solution $O = \arg \min_{S \in \Omega^T} g(S)$ with a supermodularity ratio of $\alpha(O)$, and let the greedy solution of length T be S^T . Suppose g is μ -postfix monotone with respect to O along S^T ($\mu = 1$ if g is postfix non-increasing). If $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$, then

$$g(S^T) \leq \mu (1 - \phi_T(\alpha(O))) g(O) + \phi_T(\alpha(O)) g(\emptyset), \quad \text{i.e.,} \quad g(\emptyset) - g(S^T) \geq (1 - \phi_T(\alpha(O))) (g(\emptyset) - \mu g(O)).$$

The proof of Theorem 4.1 appears in Section A.2. As $T \rightarrow \infty$, the factor $1 - \phi_T(\alpha(O))$ decreases to $1 - e^{-1/\alpha(O)}$, so the worst-case performance of the greedy rule saturates rather than degrades with horizon, provided $\alpha(O)$ and μ stay bounded as T grows (both are defined through the optimal sequence of length T and can depend on it). The bound concerns the reduction of the error over T steps relative to the best sequence of T steps: with $\alpha = \mu = 1$ it guarantees a $(1 - e^{-1})$ fraction of the optimal reduction, uniformly over ensembles, but it is not a convergence rate—a single convergent solver applied T times already reduces the error by ρ^{2T} , which for large T is far below the $e^{-1}g(\emptyset)$ term. In the settings of Section E the greedy sequence is in fact optimal or within a few percent of the exact optimum over short horizons on most instances (Section E.10), much better than the worst case. Additionally, larger $\alpha(O)$, which indicates higher reward for delayed inclusions, loosens the suboptimality bound. Theorem 4.1 requires that the sequence objective h be weakly supermodular with respect to the optimal solution and postfix monotone—properties established in Proposition 4.2. We defer the proof to Section A.3.

Proposition 4.2. Suppose that for all $j \in [K]$, the error propagation function $I - C_j \circ \mathcal{L}_h^a$ is ρ_j -Lipschitz continuous with $\rho_j \leq 1$, that $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$, and that $\sum_{i=1}^T (1 - \rho_{O_i}^2) > 0$, i.e., at least one action of the optimal sequence O is a strict contraction. Then, the function h is weakly supermodular with respect to the optimal solution O , with

$$\alpha(O) = \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

Furthermore, if the error propagation maps are linear and simultaneously diagonalizable by a unitary matrix with eigenvalues of modulus at most one (the setting of Theorem 4.3), h is also postfix non-increasing; invertibility alone does not suffice (Section A.3).

The conditions of Proposition 4.2 are natural. The corrector of Section 7 leaves the modes outside its band untouched and has $\rho = 1$ exactly; the proposition still applies because the smoother in O is a strict contraction. For classical solvers, the Lipschitz constant is $\|I_N - C_j \mathcal{L}_h^a\|$, which is typically less than 1 for well-posed linear elliptic PDEs (i.e., the update is damping errors). For neural networks, Lipschitz continuity can be enforced via weight regularization [Gouk et al., 2021] and a sufficiently trained model should approximate $(\mathcal{L}_h^a)^{-1}$ well enough to make the Lipschitz constant small. The requirement $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$ is both natural and desirable: it precludes spurious updates when the residual $f_h - \mathcal{L}_h^a u_h^{(t)}$ is 0. This holds by design for classical schemes, and it can be enforced for any learned model by excluding bias terms.

The form of $\alpha(O)$ in Proposition 4.2 highlights that the suboptimality factor in Theorem 4.1 is governed by the collective contraction factors of the solvers chosen by the optimal solution: as $\sum_{i=1}^T \rho_{O_i}^2 \rightarrow T$, $\alpha(O)$ grows, resulting in a weaker bound. Postfix monotonicity ($\mu = 1$) holds when the error propagation maps share a unitary eigenbasis—constant-coefficient periodic operators with Jacobi-type smoothers and band-limited spectral correctors—but it can fail for non-commuting contractions even when all of them are invertible, and a Gauss–Seidel sweep or a learned corrector need not be invertible at all. For the non-commuting solvers of our experiments no a priori bound on μ is available; Section E.10 computes μ , $\alpha(O)$ and the bound of Theorem 4.1 by exhaustive enumeration for short horizons ($T \leq 4$ macro-actions) and finds $\mu \leq [\text{pending}]$ and $\alpha(O) = 1$ on every ensemble of the study, including those with Gauss–Seidel, SOR and multigrid; at the horizons of tens to hundreds of macro-actions at which the oracle actually runs, only the worst-case constant of Proposition 4.2 is available, and it is loose.

Theorem 4.1 thus indicates that the approximation guarantee is strongest when the sequence is nearly supermodular ($\alpha(O) \approx 1$). Generally, if all error propagation maps share an eigenbasis, supermodularity is established. This occurs, for example, for linear, constant-coefficient PDEs with periodic boundary conditions when the solver ensemble consists of (damped) Jacobi iterations and

band-limited spectral correctors such as a single-layer linear Fourier neural operator, all of which are diagonal in the Fourier basis; lexicographic Gauss–Seidel and SOR are not, because their sweep order breaks translation invariance. The proof of Theorem 4.3 is deferred to Section A.5.

Proposition 4.3. *Let $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$ for all $j \in [K]$ and $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$ with a common unitary matrix P (orthogonal in the real symmetric case; the Jacobi matrix of the convection–diffusion operator is a real circulant with complex eigenvalues, hence unitarily but not orthogonally diagonalisable). Then, h is supermodular.*

Cost-aware greedy routing. Algorithm 1 treats every update as equally expensive, whereas in practice a neural-operator evaluation and a classical sweep have very different costs, and the quantity of interest is the error reached per unit of wall-clock time. We make the greedy rule cost-aware by changing its action set, not its comparison: let c_j denote the (measured) cost of one application of C_j , take the cost u of one neural-operator call as the unit, and replace each preconditioner by the macro-action $C_j^{(m_j)}$ that applies C_j for $m_j = \max\{1, \text{round}(u/c_j)\}$ consecutive iterations, so that every action not dearer than the unit costs u up to rounding ($m_j c_j / u$ within 0.5/ m_j of 1). For a linear update, m consecutive sweeps are again a stationary iteration, $(I - C_j \circ \mathcal{L}_h^a)^m = I - C_j^{(m)} \mathcal{L}_h^a$ with $C_j^{(m)} = \sum_{i=0}^{m-1} (I - C_j \mathcal{L}_h^a)^i C_j$, and for a nonlinear C_j the macro-action is the m -fold composition of its error propagation map. Hence $\{C_j^{(m_j)}\}_{j=1}^K$ is again a set of preconditioning functions in the sense of Section 4: if $I - C_j \circ \mathcal{L}_h^a$ is ρ_j -Lipschitz and zero-preserving, then so is its m_j -fold composition (with constant $\rho_j^{m_j}$), a shared eigenbasis is inherited by powers, and Theorem 4.1, Propositions 4.2 and 4.3, and the consistency result of Section 5 apply verbatim to an ensemble of macro-actions, all of which are equalised to the unit cost. The deployed rule is exactly Algorithm 1 on these macro-actions—the plain comparison of the errors after one macro-action—and a horizon of T macro-actions corresponds to a wall-clock budget of $\approx Tu$; it selects, at every decision epoch, the action with the largest error reduction per unit cost up to the rounding of the m_j . The one exception is an action dearer than the unit ($c_j > u$, hence $m_j = 1$), whose error reduction is compared per unit of cost, $(\|e_j\|/\|e^{(t)}\|)^{u/c_j}$: this is a geometric-rate heuristic rather than an equal-cost step, and the analysis does not cover it. With the compiled kernels of Section E every point smoother is cheaper than the unit on every grid and a multigrid cycle is cheaper than the unit at 128^2 and 256^2 , so the heuristic is only used for multigrid at 512^2 (and, in the granularity ablation of Section E.6, for the corrector itself); every other cell of the study runs Algorithm 1 on macro-actions, which is also the rule that the exhaustive check of Section E.10 verifies. (An earlier version applied the exponent $u/(m_j c_j)$ to every action, including the rounding error of the cheap ones; that rule is not Algorithm 1, and it was changed so that the theory covers what is deployed.) Section E uses this cost-aware instantiation throughout.

5 Approximate Greedy Router

The results in Section 4 indicate that the error reduction from the greedy solution closely matches that of the optimal sequence, but this is predicated on having access to the initial error, $e_h^{(0)}$. Inaccurate error estimates can result in poor solver decisions, causing errors to amplify. To remedy this, we learn a router r that selects solvers myopically, as in Algorithm 1, without access to true errors.

We adopt the following learning setup. Let $\mathcal{A}, \mathcal{F}, \mathcal{U} \subseteq \mathbb{R}^N$ denote the spaces of coefficient, forcing, and solution functions on the grid $G_h(D)$. We assume an application-specific data distribution $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$ over $\mathcal{A} \times \mathcal{F}$ that reflects test time conditions. During training, (a_h, f_h) is drawn from $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$, and a high-accuracy reference solution u_h is computed, providing the true per-step error. The router r is learned offline on this data; at test time, it operates without access to true errors.

At iteration t , router r selects a solver using the coefficient $a_h \in \mathcal{A}$, forcing $f_h \in \mathcal{F}$, and the current iterate $u_h^{(t)} \in \mathcal{U}$. If $r(a_h, f_h, u_h^{(t)}) = j$, the next iterate is computed with solver j , resulting in an error of $\|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$. Learning such a router requires minimizing the following loss:

$$l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) = \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)})=j} \quad (8)$$

with risk $\mathcal{R}_{\text{route}}(r) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} [l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h)]$. For analysis, we fix the iterate generation via teacher-forcing [Williams and Zipser, 1989, Lamb et al., 2016]: during training, the iterates fed to the router are produced by an oracle greedy rollout, not by the router’s own past choices. Formally, with $u_h^{(0)} = 0_N$, for $t \geq 1$,

$$j_t^* \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) \left(e_h^{(t-1)} \right) \right\|_2^2, \quad u_h^{(t)} = u_h^{(t-1)} + C_{j_t^*} \left(f_h - \mathcal{L}_h^a u_h^{(t-1)} \right)$$

Thus, $\{u_h^{(t)}\}$ is a deterministic function of (a_h, f_h) . At each t , l_{route} is evaluated on $r(a_h, f_h, u_h^{(t)})$ which takes in the teacher-forced iterate. Finally, the greedy choice j_{t+1}^* is used to advance $u_h^{(t+1)}$. Thus the input distribution seen by r depends on (a_h, f_h) , not on the router’s predictions. However, full teacher forcing induces a distributional mismatch at test time (exposure bias). In experiments, we mitigate this with scheduled sampling [Bengio et al., 2015]; see Section C for details.

Minimizing $\mathcal{R}_{\text{route}}(r)$ yields the following Bayes-Optimal Router:

$$r^*(a_h, f_h, u_h^{(t)}) \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) \left(e_h^{(t)} \right) \right\|_2^2 \quad (9)$$

which aligns with the update in Algorithm 1. Hence, l_{route} is consistent with learning the greedy rule.

5.1 Surrogate Loss

Since Equation (8) is discontinuous and non-convex, direct minimization is intractable in practice. We instead introduce a surrogate loss that (1) is convex, enabling efficient optimization; (2) upper bounds the original loss; and (3) is Bayes consistent, ensuring the Bayes optimal decision of the original loss is preserved upon minimization. Formally, a surrogate ϕ is considered to be Bayes consistent with respect to the loss l if

$$\lim_{n \rightarrow \infty} \mathcal{R}_\phi(f_n) - \mathcal{R}_\phi^* = 0 \implies \lim_{n \rightarrow \infty} \mathcal{R}_l(f_n) - \mathcal{R}_l^* = 0$$

where $\mathcal{R}_l(f) = \mathbb{E} [l(f(X), Y)]$ and $\mathcal{R}_l^* = \inf_f \mathcal{R}_l(f)$. In other words, in the limit of infinite data, if the risk of a sequence of learned hypotheses $\{f_n\}$ converges to the optimal risk under ϕ , it also converges to the optimal risk with respect to the original loss l .

To define a surrogate loss for the routing problem, consider a set of scoring functions $\mathbf{g} = \{g_j\}_{j=1}^K$ with $g_j : \mathcal{A} \times \mathcal{F} \times \mathcal{U} \rightarrow \mathbb{R}$ and we define the router as $r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$. For example, $\mathbf{g}$ can be a neural network with K outputs. Then, minimizing the following surrogate loss yields the same decision as minimizing Equation (8):

$$\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h) = - \sum_{j=1}^K \sum_{k=1}^K \tilde{c}_k(a_h, u_h^{(t)}, u_h) \mathbf{1}_{k \neq j} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{m=1}^K \exp(g_m(a_h, f_h, u_h^{(t)}))} \right) \quad (10)$$

where $\tilde{c}_j(a_h, u_h^{(t)}, u_h) = \|(I - C_j \circ \mathcal{L}_h^a)(u_h - u_h^{(t)})\|_2^2$. The Ψ -risk is denoted by $\mathcal{R}_\Psi(\mathbf{g}) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} [\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)]$. The convexity of Ψ with respect to the score vector $\mathbf{g}$ follows from the convexity of the negative log-softmax function in its inputs (the surrogate is convex in the scores, not in the parameters of a network that produces them). Moreover, Ψ upper bounds l_{route} up to a constant factor, and we refer the reader to Section B.2 for the proof. Finally, Theorem 5.1 shows that Ψ achieves Bayes consistency with respect to l_{route} ; the proof can be found in Section B.3.

Theorem 5.1. *Let the per-step costs be normalised state by state, $\hat{c}_j = \tilde{c}_j / \sum_{k=1}^K \tilde{c}_k$, at every state at which $\sum_k \tilde{c}_k > 0$, i.e., at which at least one solver does not annihilate the error (states at which every solver annihilates it carry no routing loss and are discarded), and let $\hat{l}_{\text{route}}$ and $\hat{\Psi}$ denote the routing loss and the surrogate with $\tilde{c}_j$ replaced by $\hat{c}_j$; the normalisation does not change the Bayes-optimal decision equation 9 at any state. Then, for any collection of solvers $\{C_j\}_{j=1}^K$ and linear discrete operator $\mathcal{L}_h^a$, $\hat{\Psi}$ is a Bayes consistent surrogate for $\hat{l}_{\text{route}}$.*

Theorem 5.1 is a cost-sensitive analogue of the classical Bayes-consistency of multiclass cross-entropy for 0-1 loss: in the infinite-sample limit, minimizing cross-entropy recovers the true conditional

class probabilities, so the induced decision is Bayes optimal. Our result extends this to cross-entropy with instance-dependent weights $\sum_{k \neq j}^K \hat{c}_k(a_h, u_h^{(t)}, u_h)$. The normalisation is what the router is trained with (Section E.1) and what makes the calibration constants meaningful: an earlier version assumed uniform absolute bounds $E_{\min} \leq \max_j \tilde{c}_j \leq \bar{E}$, which cannot hold along a convergent solve, where every $\tilde{c}_j \rightarrow 0$; with the normalised weights the only hypothesis is that at least one solver leaves a positive fraction of the error at every visited state, which Section E.10 checks. Under this condition, minimizing $\hat{\Psi}$ recovers the Bayes-optimal router of Equation (9), i.e., the greedy solution of Equation (5). Following the work of Mao et al. [2024], the proof uses standard conditional-risk calibration: we relate the excess risks of l_{route} and Ψ and take the infinite-sample limit. This is a per-decision statement on the state distribution the router is trained on: it says that minimising Ψ recovers the greedy decision at each visited state, not that the induced sequence, its time to tolerance, or its distance to the optimal sequence are controlled; the latter would require a sequential argument about the router’s own state distribution, which we do not provide and mitigate empirically with DAGger (Section E). The deployed router, moreover, does not see $(a_h, f_h, u_h^{(t)})$ but a few scalar features of the residual history (Section E.1); with respect to those features the Bayes-optimal router minimises the conditional expectation of the normalised costs given the features, which can differ from the full-information greedy decision when states with different error spectra share the same features, and the normalisation preserves the decision state by state but not necessarily after that averaging. Theorem 5.1 therefore concerns the full-information router, and the price of the feature compression is measured rather than proved, by the agreement between the deployed router and the oracle (Section E).

6 Related Works

Hybrid PDE Solvers: Early data-driven solvers sought convergence guarantees by predicting parameters—e.g., preconditioning matrices, multi-grid smoothers or restriction matrices—within iterative schemes [Taghibakhshi et al., 2021, Caldana et al., 2024, Kopaničáková and Karniadakis, 2025, Huang et al., 2022, Katrutsa et al., 2020]. These works, however, do not leverage the neural surrogates’ ability to generalize across varying coefficients and forcings highlighted in Section 2.2 and thus, offer only modest speedups over classical solvers. HINTS [Zhang et al., 2024] introduced hybrid solvers that interleave a classical method with a pre-trained DeepONet on a fixed schedule (e.g., 24 Jacobi steps, then one DeepONet correction), achieving faster convergence than the standalone numerical solver. Subsequent works extend this idea to new geometries [Kahana et al., 2023] and analyze error mode damping, replacing DeepONet with alternatives such as MIONet [Jin et al., 2022] [Hu and Jin, 2025] or FNO [Li et al., 2020a, Cui et al., 2022]. However, these hybrid solvers are limited in two ways: they, firstly, only combine the trained surrogate with a *single* numerical solver, and secondly rely on a fixed, heuristic schedule. Our proposed method addresses these shortcomings, resulting in significant empirical improvements (Section 7).

Model Routing: Routing [Shnitzer et al., 2023, Hu et al., 2024, Ding et al., 2024, Huang et al., 2025] selects, for each input, a model from a fixed set to optimize a task metric under cost or latency constraints. Simple heuristics—e.g., thresholding a cheap model’s uncertainty estimates [Chuang et al., 2024, 2025]—often poorly balance the cost-accuracy tradeoff, motivating many systems to instead learn a router that maps inputs to the best model [Hari and Thomson, 2023, Mohammadshahi et al., 2024, Šakota et al., 2024]. We similarly learn a router, but over numerical and neural solvers for PDEs at the iteration level. Our work is the first to apply routing to the problem of learning hybrid PDE solvers, in contrast to prior approaches that use fixed schedules. Additionally, by exploiting the algebraic structure of PDE solvers, we derive theoretical guarantees for our routing strategy (Section 4), unlike typical model-routing settings.

Mixture of Experts: Classical mixture-of-experts (MoE) [Jacobs et al., 1991, Jordan and Jacobs, 1994] uses a gating network to combine the outputs of multiple experts, trained jointly with the gate. In modern LLMs, MoE instead performs sparse, token-level routing to a subset of in-layer experts, enabling large capacity without proportional compute, typically with load-balancing mechanisms [Shazeer et al., 2017, Lepikhin et al., 2020, Fedus et al., 2022, Jiang et al., 2024]. Our method can be regarded as gating of a different kind: “experts” (solvers) are not jointly trained with the router, and a single router is reused across iterations, unlike MoE which commonly employs layer-specific gates.

7 Experiments

We evaluate the approximate greedy router on four PDEs posed on the unit domain $D = [0, 1]^2$ with periodic boundary conditions:

Poisson: $-\Delta u = f$, **ConvDiff:** $-\Delta u + 20 \partial_{x_1} u + 20 \partial_{x_2} u = f$, **AnisoDiff:** $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$, $\epsilon = 10^{-2}$, **VarCoef**

where f is centred to satisfy the compatibility condition $\int_D f(\mathbf{x}) d\mathbf{x} = 0$ and is drawn from a zero-mean hierarchical Gaussian random field with covariance operator $\alpha(-\Delta + \beta I)^{-\gamma}$ and sample-dependent (α, β, γ) (Section C). Anisotropic diffusion is the standard problem on which point relaxation fails to smooth: an error mode oscillatory in x_1 and smooth in x_2 is damped by only $(1 - \epsilon)/(1 + \epsilon) \approx 0.98$ per Jacobi sweep. Variable-coefficient diffusion uses a fixed smooth coefficient field $a(\mathbf{x}) = \exp(\kappa z(\mathbf{x}))$ with $\max a / \min a = 10$: its discrete operator is not diagonalised by the FFT, so no direct spectral solve exists, the corrector’s band-limited inverse has no closed form and is fitted, and the fitted corrector is only approximately accurate (Section E.5). The domain is discretised on 128^2 , 256^2 and 512^2 grids; Dirichlet boundary conditions are not covered by this study (Section D). Because iteration counts favour methods whose iterations are expensive, our primary metric is the end-to-end wall-clock time to a target relative error $\varepsilon = \|e_h^{(t)}\| / \|u_h\|$ on 64, 32 and 16 test instances per grid, with $\varepsilon = h^2$ —a grid-dependent tolerance that lies below the measured discretisation error of every test instance, so that further algebraic refinement is very unlikely to improve the computed solution (Section E.1)—highlighted; every comparison is paired over instances and tested with a two-sided Wilcoxon signed-rank test with a Holm correction. We also report the iteration-based metric of the original submission, the area under the error curve, $\text{AUC} = \sum_{t=1}^T \|e_h^{(t)}\| / \|u_h\|$ over $T = 300$ iterations. Section E gives the full protocol and collects every result per equation.

Setup. The classical solvers are Jacobi, damped Jacobi ($\omega = 0.67$), Gauss–Seidel (GS), symmetric GS (SymGS), SOR ($\omega = 1.5$), line Gauss–Seidel (anisotropic diffusion only: every grid line along the strongly coupled axis solved exactly, the classical remedy for anisotropy) and geometric multigrid (V(2,2)-cycles with GS smoothing; line-GS smoothing for anisotropic diffusion), all implemented as compiled single-threaded C stencil kernels (a GS sweep costs [pending] and a V-cycle [pending] at 128^2 , measured in the form in which the benchmark charges them), so the classical baselines are not slowed by the implementation. The neural operator is a DeepONet corrector that acts on the residual through a 64×64 sensor grid with a Fourier trunk basis and is applied scale-equivariantly: its deployed form is a linear, Fourier-structured map (a least-squares fit of the band-limited inverse of the discrete operator, which it reproduces to $\approx 10^{-6}$ on the constant-coefficient equations), so one call solves the smooth band of the spectrum at the cost of about [pending] Jacobi or [pending] GS sweeps at 128^2 (and [pending] Jacobi sweeps at 512^2 , since its cost is nearly grid-independent) and never injects error outside its band. The router is made *cost-aware* through the cost-equalised macro-actions of Section 4 and selects from $6+K$ scalar features (log residual, its recent decay rate, the previous action and call counts) with a two-layer MLP whose decision costs [pending]–[pending] μs in situ, independently of N ; it is trained with the surrogate of Section 5.1 to imitate the cost-aware greedy oracle on oracle rollouts followed by DAgger rounds. We compare against the classical solver alone; HINTS with the period $\tau=15$ that Zhang et al. [2024] use in all of their two-dimensional experiments ($\tau=25$, their one-dimensional period, is reported in Section E); the best fixed schedule chosen *a posteriori* on the test set for each cell from HINTS with $\tau \in \{2, 5, 10, 15, 25, 50\}$, phase-shifted HINTS (first call at iteration 0) and the static one-shot schedule (corrector once at iteration 0, then the solver); the best of three residual-decay rules—deterministic controls at the router’s granularity that call the corrector again whenever the residual’s contraction over the last macro-action falls below a threshold, also chosen on the test set; the cost-aware greedy oracle (Algorithm 1 with the true error; not deployable); and the classical methods a practitioner would use on these problems without a corrector—multigrid alone (line-smoothed for anisotropic diffusion), multigrid-preconditioned CG or BiCGSTAB, and the FFT or sparse direct solve—timed inside the same replay loop as the policies. The forcings are sampled from the hierarchical GRF with the Hermitian symmetry of a real field and without Nyquist modes (Section E.1). All configurations were fixed on a development test set, and every number reported is a single confirmatory evaluation on a fresh test set with three timed replays in random order and a drift guard per instance (Section E.9); paired comparisons use two-sided Wilcoxon tests with a Holm correction and bootstrap intervals.

Paired solvers. Table 1 reports, for every pairing $\{\text{NO}, \text{solver}\}$, equation and grid, the median wall-clock time to $\varepsilon = h^2$ of HINTS, of the best fixed schedule, and of the learned router, together

with the classical methods without corrector. Two regimes appear. *At truncation level* the whole solve is one corrector call followed by a few sweeps: the router, the cost-aware oracle and the best static schedule (the corrector at iteration 0, then smoothing) take the same decisions, so the router is within a few percent of the best fixed schedule found by a test-set search—paired ratios [pending]–[pending] at 128^2 , [pending]–[pending] at 256^2 and [pending]–[pending] at 512^2 , the difference being its own decision overhead—without any search, and it is [pending]–[pending] faster than HINTS with its published period $\tau=15$ ([pending]–[pending] at 256^2 , [pending]–[pending] at 512^2 ; [pending]–[pending] against $\tau=25$), whose first call comes $\tau - 1$ sweeps too late; the best residual-decay rule, which also calls the corrector first, is within [pending]–[pending] of the router there; the classical solver alone is [pending]–[pending] slower on the isotropic equations and never reaches h^2 on the anisotropic one. *Below truncation level* the schedule matters and the picture is mixed: down to $\varepsilon = 10^{-8}$ the router is [pending]–[pending] faster than HINTS ($\tau=25$) and one to two orders of magnitude faster than the one-shot schedule, because it calls the corrector again when the smooth error has re-emerged (Figure 1), but it beats the best fixed schedule in only [pending] of the [pending] pairings at 128^2 (paired ratios [pending]–[pending]) and the best residual-decay rule in [pending] ([pending]–[pending]): the best static schedule there is a phase-shifted HINTS with a short period, which calls the corrector every 5–10 sweeps, more often than a cost-equalised macro-action of [pending] Gauss–Seidel or [pending] Jacobi sweeps allows the router to (Section E.6 examines this granularity limit). The router tracks the cost-aware oracle within [pending] in median time; the stability of its decisions over five independent training seeds is reported in Section E [pending]. Against the classical methods without corrector, the compiled multigrid is a strong competitor: at 128^2 the stationary hybrids match it at h^2 (router [pending]–[pending] of a stand-alone V-cycle solve) and are slower than it below h^2 ([pending]–[pending]), whereas routing over {NO, multigrid} is [pending]–[pending] and [pending]–[pending] of multigrid alone at h^2 and 10^{-8} ; the router is [pending]–[pending] of multigrid-preconditioned CG or BiCGSTAB at h^2 , and on the constant-coefficient equations the FFT direct solve remains [pending]–[pending] faster than any iterative method including ours, whereas on variable-coefficient diffusion, which has no such solve, the router is compared with the sparse direct solve (Section E.5). The iteration-based metrics tell the same story: at 128^2 the router’s AUC over $T = 300$ iterations is orders of magnitude below HINTS’ and the solver-only baseline’s for every pairing and equation (paired t -tests on log AUC, $p < 10^{-3}$; Tables 8, 14, 20 and 26).

Solver ensembles. The same construction routes over ensembles $\text{NO} \cup \mathcal{W}$ of several classical solvers. Since the greedy rule is a per-decision comparison, a larger ensemble can only help where the error passes through regimes that favour *different* members; an exhaustive oracle-level screening of every subset of up to three members (Section E.7) shows that this never happens at $\varepsilon = h^2$ —one corrector call and one macro-action of the cheapest smoother already reach truncation level—and, in the development runs, happened at 10^{-8} through a nonstationary Jacobi schedule (undamped Jacobi on the band-edge modes, damped Jacobi on the near-Nyquist modes). We evaluate the learned router over the nested sets {Jacobi}, {Jacobi (0.67)}, {Jacobi, Jacobi (0.67)} and {Jacobi, Jacobi (0.67), GS} and the four larger sets of Table 30 on every equation, with the single-member routers evaluated in the same session on the same instances. At 128^2 the confirmatory run shows no separation: at $\varepsilon = 10^{-8}$ the ensemble router is significantly faster than the best single-member router in [pending] of [pending] ensemble cells, and the ensemble *oracle* is itself [pending]–[pending] of the best single-member oracle, because with the compiled kernels a macro-action is about forty Jacobi sweeps and a single damped-Jacobi macro-action already removes the whole band the corrector leaves at 128^2 ; the nonstationary schedule of the development runs, where a macro-action was about ten sweeps, has nothing left to exploit. Whether the gain returns at a finer decision granularity (Section E.6) and on the 256^2 grids, where the band is twice as wide, is [pending]. Enlarging the set is free at h^2 (every ensemble router within [pending]–[pending] of the best single-member router in work units), whereas at 10^{-8} the ensemble routers are [pending]–[pending] of it: the greedy oracle is not monotone in its action set, and its imitation degrades among near-tied members (Section E.7). Finally, we verify numerically that the assumptions of Sections 4 and 5 hold in all these settings, in the energy norm for the non-normal sweeps (Section E.10).

Every result per equation and grid, the ensemble study, the accounting of overheads and training cost, and the verification of the theoretical assumptions are collected in Section E; Section D reports the Dirichlet experiment of the original submission.

Table 1: Median wall-clock time to truncation-level relative error $\varepsilon = h^2$ for every solver pairing $\{\text{NO}, \text{solver}\}$, equation and grid ($64, 32$ and 16 test instances at $128^2, 256^2$ and 512^2): HINTS with the published period $\tau=25$, HINTS with the best period chosen per cell on the test set itself (subscript), and the learned router (paired median speedup over HINTS-25 / over the best period in parentheses; bold: faster than both). The classical solver alone is $10^2\text{--}10^3\times$ slower than the router on the isotropic equations and never reaches h^2 on the anisotropic one (Section E). The last rows of each block give geometric multigrid alone and the best multigrid-preconditioned Krylov method, without corrector, on the same instances. Anisotropic diffusion is not run at 512^2 : its corrector keeps full resolution along x_1 , and the resulting $65,536 \times 65,536$ branch matrix does not fit in memory (Section E); multigrid is not run on it either (Section E.1). $\dagger n$: n instances did not reach h^2 within the iteration cap.



Figure 1: Corrector usage on the 128×128 grid: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ($\tau=25$), for every solver pairing (columns) and all three equations (rows); the same figure for the 256^2 and 512^2 grids is in Section E.

8 Discussion

We have introduced an adaptive method for selecting solvers that efficiently minimizes the final error of a PDE in iterative methods, opening many directions for future work. Our current DeepONet is trained without accounting for its downstream role in correction term prediction; jointly training the ML solver and router could yield larger gains. Another avenue is to use reinforcement learning to learn a cost-aware routing strategy that optimizes error under compute budgets; Sections 4 and E show that a myopic cost-aware variant of the greedy rule—Algorithm 1 on cost-equalised macro-actions—already reproduces the best static schedule at truncation level and gives mixed results below it, and RL could extend it to non-myopic budgeted routing. RL would additionally enable extending the ensemble routing to *continuous* parameterization of solvers. Finally, framing routing as an online learning problem could enable adaptation to unknown deployment conditions.

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712 A Proofs for Section 4

713 A.1 Proof of Proposition A.1

714 **Proposition A.1.** *Any prefix monotonically non-increasing sequence supermodular function g is*
715 *weakly supermodular with respect to all sequences $S \in \Omega^*$ with $\alpha(S) = 1$*

716 *Proof.* This proof is adapted from Liberty and Sviridenko [2017].

717 If g is sequence supermodular then,

$$\begin{aligned} & g(S) - g(S \oplus S') \\ &= \sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) - g(S \oplus (S'_1, \dots, S'_i)) \\ &\stackrel{(a)}{\leq} \sum_{i=1}^{|S'|} g(S) - g(S \oplus S'_i) \\ &\leq |S'| \max_{i \in [|S'|]} g(S) - g(S \oplus S'_i) \end{aligned}$$

718 (a) by supermodularity

719 □

720 A.2 Proof of Theorem 4.1

721 **Theorem 4.1.** *Let $g : \Omega^* \rightarrow \mathbb{R}$ be a weakly supermodular function with respect to the optimal*
722 *solution $O = \arg \min_{S \in \Omega^T} g(S)$ with a supermodularity ratio of $\alpha(O)$ and μ -postfix monotonicity*
723 *with respect to O along the greedy solution S^T of length T . If $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$, then*

$$g(S^T) \leq \mu(1 - \phi_T(\alpha(O)))g(O) + \phi_T(\alpha(O))g(\emptyset)$$

724 *Proof.* This proof strategy is inspired by Streeter and Golovin [2008] and Liberty and Sviridenko
725 [2017]

$$\begin{aligned} g(S^t) - \mu g(O) &\stackrel{(a)}{\leq} g(S^t) - g(S^t \oplus O) \\ &= \sum_{i=1}^{|O|} g(S^t \oplus (o_1, \dots, o_{i-1})) - g(S^t \oplus (o_1, \dots, o_i)) \\ &\stackrel{(b)}{\leq} \alpha(O) \sum_{i=1}^{|O|} g(S^t) - g(S^t \oplus o_i) \\ &\leq \alpha(O)|O| \max_{i \in [|O|]} g(S^t) - g(S^t \oplus o_i) \\ &\leq \alpha(O)Tg(S^t) - \alpha(O)T \min_{\omega \in \Omega} g(S^t \oplus \omega) \\ &= \alpha(O)Tg(S^t) - \alpha(O)Tg(S^{t+1}) \end{aligned}$$

726 (a) by μ -postfix monotonicity along S^T , i.e., $g(S^t \oplus O) \leq \mu g(O)$; (b) by weak supermodularity with
727 respect to O (applied to the prefix S^t).

728 After rearranging the inequality, we get:

$$\begin{aligned} g(S^{t+1}) &\leq \frac{1}{\alpha(O)T} (\mu g(O) + (\alpha(O)T - 1) g(S^t)) \\ &= \frac{\mu}{\alpha(O)T} g(O) + \left(1 - \frac{1}{\alpha(O)T}\right) g(S^t) \end{aligned}$$

729 When recursively applying this inequality, we get:

$$\begin{aligned} g(S^T) &\leq \frac{\mu}{\alpha(O)T} g(O) \sum_{i=0}^{T-1} \left(1 - \frac{1}{\alpha(O)T}\right)^i + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset) \\ &= \mu \left(1 - \left(1 - \frac{1}{\alpha(O)T}\right)^T\right) g(O) + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset) \end{aligned}$$

730

□

731 A.3 Proof of Proposition 4.2

732 **Proposition 4.2.** Suppose that for all $j \in [K]$, the error propagation function $I - C_j \circ \mathcal{L}_h^a$ is
 733 ρ_j -Lipschitz continuous with $\rho_j \leq 1$, that $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$, and that $\sum_{i=1}^T (1 - \rho_{O_i}^2) > 0$,
 734 i.e., at least one action of the optimal sequence O is a strict contraction. Then, the function h is
 735 weakly supermodular with respect to the optimal solution O , with

$$\alpha(O) = \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

736 Furthermore, if the error propagation maps are linear and simultaneously diagonalizable by a
 737 unitary matrix with eigenvalues of modulus at most one (the setting of Theorem 4.3), h is also postfix
 738 non-increasing; invertibility alone does not suffice (Section A.3).

739 *Proof.* For brevity, we use the notation $(g_1 \circ \dots \circ g_T)(x) = \circ_{t=1}^T g_t(x)$, where composition is
 740 applied from right to left so that g_T acts first. In this proof, we use a few properties of Lipschitz
 741 continuous functions:

- 742 • **Property 1:** If g is ρ -Lipschitz continuous and $g(0) = 0$, then $\|g(x)\|_2 = \|g(x) - 0\|_2 =$
 743 $\|g(x) - g(0)\|_2 \leq \rho \|x - 0\|_2 = \rho \|x\|_2$
- 744 • **Property 2:** If g_1 and g_2 are Lipschitz continuous functions with Lipschitz constants of ρ_1
 745 and ρ_2 respectively, the Lipschitz constant of $g_1 + g_2$ and $g_1 - g_2$ is $\rho_1 + \rho_2$.
- 746 • **Property 3:** If g_1 and g_2 are Lipschitz continuous functions with Lipschitz constants of ρ_1
 747 and ρ_2 respectively, the Lipschitz constant of $g_1 \circ g_2$ is $\rho_1 \rho_2$.

748 In order to prove weakly- α -supermodularity, we must first prove prefix monotonicity.

749 **Prefix monotonicity:** Let $S \preceq S'$ where $S' = S \oplus N$.

$$\begin{aligned} h(S') &= \left\| \circ_{t=|S'|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &= \left\| \circ_{t=|S'|+1}^{|S'|+1} (I - C_{S'_t} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\stackrel{(a)}{\leq} \left(\prod_{t=|S|+1}^{|S'|} \rho_{S'_t}^2 \right) \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\leq h(S) \end{aligned}$$

750 (a) by Property 1

751 **Weak supermodularity:** We will upper bound $\alpha(O)$ by providing an upper bound for $h(S) - h(S \oplus$
 752 $O)$ and a lower bound for $\sum_{i=1}^T h(S) - h(S \oplus O_i)$. Since $h \geq 0$, $h(S) - h(S \oplus O) \leq h(S) =$
 753 $\left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2$. (An earlier version of this step used $\|x\|^2 - \|Gx\|^2 \leq \|x - Gx\|^2$,
 754 which is false, e.g. for $G = \frac{1}{2}I$; the direct bound is sharper by a factor 4.)

755 To lower bound $\sum_{i=1}^{|O|} h(S) - h(S \oplus O_i)$

$$\begin{aligned} \sum_{i=1}^{|O|} h(S) - h(S \oplus O_i) &= \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 - \left\| (I - C_{O_i} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\geq \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 - \rho_{O_i}^2 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &= \left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \left(T - \sum_{i=1}^T \rho_{O_i}^2 \right) \end{aligned}$$

756 If $h(S) = 0$ then, by prefix monotonicity, every term above vanishes and the weak-supermodularity
 757 inequality holds for any $\alpha \geq 1$; assume $h(S) > 0$. The lower bound just obtained is positive because
 758 $T - \sum_i \rho_{O_i}^2 > 0$ by hypothesis (this is the only place where a strict contraction is needed; $\rho_j \leq 1$
 759 suffices everywhere else). Finally,

$$\begin{aligned} \frac{h(S) - h(S \oplus O)}{\sum_{i=1}^T (h(S) - h(S \oplus O_i))} &\leq \max \left\{ \frac{\left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2}{\left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \left(T - \sum_{i=1}^T \rho_{O_i}^2 \right)}, 1 \right\} \\ &= \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\} \end{aligned}$$

760 **Postfix monotonicity.** Let $S' = S \oplus N$ and suppose that $I - C_j \mathcal{L}_h^a = P \Lambda_j P^*$ for all j with a
 761 common unitary P and $|\lambda_{ji}| \leq 1$ (the products below are of $|\lambda_{ji}|^2$ and $|z_i|^2$; we write them without
 762 moduli for the real case). By Theorem A.2,

$$h(S') = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(N)} \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \leq \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(N)} = h(N),$$

763 since every factor $\lambda_{ji}^{2m_j(S)}$ lies in $[0, 1]$. The shared eigenbasis cannot be replaced by invertibility of
 764 the individual maps: for

$$A = \begin{bmatrix} 0.0674240 & -0.1864331 \\ -0.1473150 & -0.8707252 \end{bmatrix}, \quad B = \begin{bmatrix} 0.7575738 & 0.4816284 \\ -0.1369844 & 0.3257283 \end{bmatrix}, \quad x = \begin{bmatrix} 0.2812107 \\ -0.5538228 \end{bmatrix},$$

765 both matrices are invertible with spectral norm 0.9, yet $\|BAx\|_2 = 0.330 > 0.225 = \|Bx\|_2$, i.e.,
 766 $h((A) \oplus (B)) > h((B))$. (An earlier version of this proof bounded $\|G^{-1}\|$ by the reciprocal of the
 767 Lipschitz constant of G , which is not valid.)

768 □

769 A.4 Proof of Theorem A.2

770 **Lemma A.2.** Let $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$ where P is a unitary matrix and $\Lambda_j =$
 771 $\text{diag}(\lambda_{j1}, \dots, \lambda_{jN})$. If $P^{-1} e_h^{(0)} = z$, then the following equality holds (written for a real or-
 772 thogonal P and real eigenvalues; in general z_i^2 and λ_{ji}^{2m} are to be read as $|z_i|^2$ and $|\lambda_{ji}|^{2m}$ here and

773 in the proofs of Propositions 4.2 and 4.3, which use only $\|Px\| = \|x\|$ and $|\lambda_{ji}| \leq 1$):

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \quad (11)$$

774 where $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$

Proof.

$$\begin{aligned} h(S) &= \left\| \prod_{t=|S|}^1 (I_N - C_{S_t} \mathcal{L}_h^a) e_h^{(0)} \right\|^2 \\ &= \left\| \prod_{t=|S|}^1 (P \Lambda_{S_t} P^{-1}) e_h^{(0)} \right\|^2 \\ &= \left\| P \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|^2 \\ &= \left\| \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|^2 \\ &= \sum_{i=1}^N z_i^2 \prod_{t=|S|}^1 \lambda_{S_t i}^2 \\ &= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \end{aligned}$$

775

□

776 A.5 Proof of Theorem 4.3

777 **Proposition 4.3.** Let $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$ for all $j \in [K]$ and $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$ with a common
778 unitary matrix P . Then, h is supermodular.

779 *Proof.* Let $S \preceq S'$ where $S' = S \oplus B$. By Theorem A.2,

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)}$$

780 where $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$ be the number of times a sequence S calls the solver j . Recall that h is
781 considered sequence supermodular if $\forall S', S \in \Omega^*$ such that $S \preceq S'$, it holds that

$$h(S) - h(S \oplus \omega) \geq h(S') - h(S' \oplus \omega)$$

782

$$\begin{aligned} h(S) - h(S \oplus \omega) &= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} - \sum_{i=1}^N z_i^2 \lambda_{\omega i}^2 \prod_{k=1}^K \lambda_{ki}^{2m_k(S)} \\ &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \end{aligned}$$

783 Similarly,

$$\begin{aligned}
h(S') - h(S' \oplus \omega) &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S')} \\
&\stackrel{(a)}{=} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2(m_j(S) + m_j(B))} \\
&\stackrel{(b)}{\leq} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \\
&= h(S) - h(S \oplus \omega)
\end{aligned}$$

784 (a) Since $m_j(S') = \sum_{t=1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S'_t=j} + \sum_{t=|S|+1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=k} + \sum_{t=1}^{|B|} \mathbf{1}_{B_t=j} =$
785 $m_j(S) + m_j(B)$, (b) since $|\lambda_{ji}| \leq 1$ (as $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$ and P is unitary) and $m_j(B) \geq 0$ for
786 all $j \in [K]$

787 □

788 B Proofs for Section 5

789 B.1 Proof of Theorem B.1

790 **Lemma B.1.** *For any set of preconditioning functions $\mathcal{C}$, any discrete operator $\mathcal{L}_h^a$, any router r , any*
791 *$a_h, f_h, u_h^{(t)}, u_h \in \mathcal{A} \times \mathcal{F} \times \mathcal{U} \times \mathcal{U}$, the following equality holds true:*

$$\begin{aligned}
l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) &= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2
\end{aligned}$$

792 *Proof.* Note that $\sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} = K-1$

$$\begin{aligned}
l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) &= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) = j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + (K-1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 - (K-1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + \sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \\
&\quad - (K-1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2
\end{aligned}$$

$$\begin{aligned}
&= \sum_{j=1}^K \left(\sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 - \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \right) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K-2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \\
&= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K-2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2
\end{aligned}$$

793

□

794 B.2 Proof of Theorem B.2

795 **Proposition B.2.** For any router r defined by $r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$, any $a_h \in \mathcal{A}$,
796 $f_h \in \mathcal{F}$, and $u_h^{(t)}, u_h \in \mathcal{U}$, the routing loss l_{route} satisfies:

$$\log(2) l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) \leq \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)$$

797 *Proof.* By Theorem B.1, we know that

$$\begin{aligned}
\log(2) l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) &= \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - \log(2) (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \\
&\leq \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\stackrel{(a)}{\leq} - \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \log \left(\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \\
&= \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)
\end{aligned}$$

798 (a) if $r(a_h, f_h, u_h^{(t)}) \neq j$, $\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \leq 0.5$ (with equality only at a tie for the largest

799 score), which implies that $-\log \left(\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \geq \log(2) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j}$

800

□

801 B.3 Proof of Theorem 5.1

802 **Theorem 5.1.** Let the per-step costs be normalised state by state, $\hat{c}_j = \tilde{c}_j / \sum_{k=1}^K \tilde{c}_k$, at every state at
803 which $\sum_k \tilde{c}_k > 0$, i.e., at which at least one solver does not annihilate the error (states at which every
804 solver annihilates it carry no routing loss and are discarded), and let $\hat{l}_{\text{route}}$ and $\hat{\Psi}$ denote the routing
805 loss and the surrogate with $\tilde{c}_j$ replaced by $\hat{c}_j$; the normalisation does not change the Bayes-optimal
806 decision equation 9 at any state. Then, for any collection of solvers $\{C_j\}_{j=1}^K$ and linear discrete
807 operator $\mathcal{L}_h^a$, $\hat{\Psi}$ is a Bayes consistent surrogate for $\hat{l}_{\text{route}}$.

808 *Proof.* Throughout the proof $\tilde{c}_j$ stands for the normalised weight $\hat{c}_j$ of the statement, so that $\sum_j \tilde{c}_j =$
809 1 at every state considered; consequently $\tilde{c}_j \leq \bar{E} := 1$ for all j and $\sum_k B_k = (K-1) \sum_j \tilde{c}_j =$

810 $(K-1)E_{\min}$ with $E_{\min} := 1$. With these two constants the argument below, written for general
 811 bounds $\bar{E}$ and $E_{\min}$, is unconditional. For a given a_h, f_h , let u_h be $\mathcal{G}_h(a_h, f_h)$ where $\mathcal{G}_h$ denotes
 812 the solution operator acting on the grid G_h . Furthermore, let's consider routers of the form

$$r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$$

813 For a given $a_h, f_h, u_h^{(t)} \in \mathcal{A} \times \mathcal{F} \times \mathcal{U}$, let the optimal loss under l_{route} be
 814 $l_{\text{route}}^*(a_h, f_h, u_h^{(t)}) = \inf_{\tilde{r}} l_{\text{route}}(\tilde{r}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h))$. Similarly, let the optimal loss un-
 815 der Ψ be $\Psi^*(a_h, f_h, u_h^{(t)}) = \inf_{\tilde{\mathbf{g}}} \Psi(\tilde{\mathbf{g}}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h))$. Let $B_j(a_h, f_h, u_h^{(t)}) =$
 816 $\sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a)(\mathcal{G}_h(a_h, f_h) - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j}$.

$$\begin{aligned} & l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - l_{\text{route}}^*(a_h, f_h, u_h^{(t)}) \\ & \stackrel{(a)}{=} \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a)(u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a)(u_h - u_h^{(t)}) \right\|_2^2 \\ & \quad - \inf_{\tilde{r}} \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a)(u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} + (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a)(u_h - u_h^{(t)}) \right\|_2^2 \\ & = \sum_{j=1}^K B_j(a_h, f_h, u_h^{(t)}) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K B_j(a_h, f_h, u_h^{(t)}) \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \\ & = \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \left(\sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \right) \end{aligned}$$

817 (a) by Theorem B.1

818 Let $\mathcal{X} = \mathcal{A} \times \mathcal{F} \times \mathcal{U}$ and $\mathcal{Y} = [K]$. Let $\mathcal{P}_{\mathcal{X}}$ denote the degenerate distribution supported at the
 819 point $(a_h, f_h, u_h^{(t)})$. We define the conditional distribution - $P(Y = j \mid X = (a_h, f_h, u_h^{(t)})) =$
 820 $\frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})}$ for $j \in [K]$. The risk and optimal risk of 0-1 loss under this distribution can
 821 be written as:

$$\begin{aligned} \mathcal{R}_{0-1}(r) &= \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\ \mathcal{R}_{0-1}^* &= \inf_r \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \end{aligned}$$

822 If $r(a_h, f_h, u_h^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a_h, f_h, u_h^{(t)})$ for all $x \in \mathcal{X}$, then the risk and optimal risk of
 823 cross entropy loss $(l_{ce}(\mathbf{g}, x, y) - \log(\frac{\exp(g_y(x))}{\sum_{k=1}^K \exp(g_k(x))}))$ under this distribution can be written as:

$$\begin{aligned} \mathcal{R}_{ce}(\mathbf{g}) &= - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \\ \mathcal{R}_{ce}^* &= \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \end{aligned}$$

824 From Theorem 3.1 of [Mao et al., 2023], $\mathcal{R}_{0-1}(r) - \mathcal{R}_{0-1}^* \leq \Gamma^{-1}(\mathcal{R}_{ce}(\mathbf{g}) - \mathcal{R}_{ce}^*)$ if
 825 $r(a_h, f_h, u_h^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a_h, f_h, u_h^{(t)})$ where $\Gamma(z) = \frac{1+z}{2} \log(1+z) + \frac{1-z}{2} \log(1-z)$.
 826 Then,

$$\begin{aligned}
& l_{\text{route}} \left(r, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h) \right) - l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \\
&= \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \left(\sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \right) \\
&\leq \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&\stackrel{(a)}{\leq} \bar{E} K (K-1) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&\stackrel{(b)}{\leq} \bar{E} K (K-1) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&= \bar{E} K (K-1) \Gamma^{-1} \left(\frac{\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \right)
\end{aligned}$$

827 (a) since $\left\| (I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)}) \right\|_2^2 < \bar{E}$ for all $j \in [K]$, (b) since Γ^{-1} is non-decreasing and

828 $\exists j \in [K]$ such that $\left\| (I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)}) \right\|_2^2 > E_{\min}$

829 Finally,

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \mathcal{R}_{\text{route}}(r_n) - \mathcal{R}_{\text{route}}^* \\
&\stackrel{(a)}{=} \lim_{n \rightarrow \infty} \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[l_{\text{route}} \left(r_n, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h) \right) - l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \right] \\
&\leq \lim_{n \rightarrow \infty} \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[\bar{E} K (K-1) \Gamma^{-1} \left(\frac{\Psi(\tilde{\mathbf{g}}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \right) \right] \\
&\stackrel{(b)}{\leq} \lim_{n \rightarrow \infty} \bar{E} K (K-1) \Gamma^{-1} \left(\frac{\mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[\Psi(\mathbf{g}_n, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)}) \right]}{(K-1)E_{\min}} \right)
\end{aligned}$$

830

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \bar{E} K (K - 1) \Gamma^{-1} \left(\frac{\mathcal{R}_\Psi(\mathbf{g}_n) - \mathcal{R}_\Psi^*}{(K - 1)E_{min}} \right) \\
&\stackrel{(c)}{=} \bar{E} K (K - 1) \Gamma^{-1} \left(\frac{\lim_{n \rightarrow \infty} \mathcal{R}_\Psi(\mathbf{g}_n) - \mathcal{R}_\Psi^*}{(K - 1)E_{min}} \right) \\
&= \bar{E} K (K - 1) \Gamma^{-1}(0) \\
&\stackrel{(d)}{=} 0
\end{aligned}$$

831 (a) $\mathcal{R}_{\text{route}}^* = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \right]$ since the infimum is taken over all measurable
832 functions, (b) by Jensen’s inequality since Γ^{-1} is concave, (c) by continuity of Γ^{-1} at 0, (d) $\Gamma^{-1}(0) =$
833 0

834

□

835 C Training Details

836 The periodic experiments of Section 7 use the corrector and the lightweight router described in
837 Section E.1; the data distribution below is shared by all experiments, and the DeepONet and LSTM
838 router described afterwards are those of the original submission’s pipeline, whose Dirichlet experiment
839 has been withdrawn (Section D); they are kept for the record.

840 Data for both DeepONet and the routers is sampled from a zero-mean Hierarchical Gaussian Random
841 Field on the periodic domain with covariance operator $\alpha(-\Delta + \beta I)^{-\gamma}$, where α, β, γ are sampled
842 as:

$$\begin{aligned}
\alpha &\sim \text{Log-Uniform}(0.01, 100) \\
\beta &\sim \text{Log-Uniform}(0.1, 1000) \\
\gamma &\sim \text{Uniform}(\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0\})
\end{aligned}$$

843 Given (α, β, γ) , samples are generated in the Fourier space. For each non-zero frequency mode
844 k , we draw an independent complex coefficient from a Gaussian distribution with mean 0 and
845 variance $\alpha(4\pi^2\|k\|_2^2 + \beta)^{-\gamma}$. We enforce a Hermitian symmetry to obtain a real-valued field, set the
846 zero-frequency (DC) mode to 0 to obtain a zero mean field, and apply the inverse Discrete Fourier
847 Transform to obtain the field in physical space.

848 For each sample, we compute reference solutions with a least squares solver and treat them as ground
849 truth.

850 This data is used to trained our DeepONet models and LSTM routers. All the models were imple-
851 mented using PyTorch and all the models were trained on one Nvidia A40 GPU.

852 Table 2 contains all hyperparameter details for the DeepONet. DeepONet took 30 minutes to train.
853 We then use the model with the best validation loss.

Table 2: Hyperparameter settings for DeepONet

Hyperparameter	Value
Learning rate	1e-3
Branch Dimension	128
Hidden dimension for branch net	256
No. of hidden layers in branch net	4
Hidden dimension for trunk net	256
No. of hidden layers in trunk net	4
Gradient Clipping Norm	1.0
Weight Decay	0.005
Batch size	256
Training samples	10000
Validation samples	2000
Epochs	1000

854 The routers are LSTM models. Along with the inputs specified in Section 5, namely $a_h, f_h, u_h^{(t)}$,
855 we also supply the routers with the iteration index t , current residual $f_h - \mathcal{L}_h^a u_h^{(t)}$, and the routing
856 decisions from the previous iteration. While the iteration index and residual are deterministic
857 functions of $(a_h, f_h, u_h^{(t)})$ and align with the theory presented in Section 5, incorporating previous
858 routing decisions is a mild deviation from the idealized setting. Nevertheless, this modification is
859 motivated by the same considerations that justify the use of scheduled sampling: under deployment,
860 routing decisions are predicted autoregressively and influence future inputs, whereas training under
861 teacher forcing assumes independence from past predictions. Including previous routing decisions
862 bridges the gap induced by exposure bias by providing the model with trajectory-level information.

863 All the routers are trained with scheduled sampling. We use a warm-up of e_w epochs with teacher-
864 forcing probability $p_{tf}(e) = ss_{start}$. After the warm-up, the p_{tf} decays geometrically by a factor of
865 $\gamma_{tf} < 1$ per epoch and is floored by ss_{end} :

$$p_{tf}(e) = \begin{cases} ss_{start} & e \leq e_w \\ \max(ss_{start}\gamma_{tf}^{e-e_w}, ss_{end}) & e > e_w \end{cases}$$

866 At each time step, with probability $p_{tf}(e)$, we feed the teacher-forced greedy iterate; otherwise, we
867 feed the router’s own predicted iterate.

868 Since LSTMs on long rollouts can suffer from exploding/vanishing gradients, we use truncated
869 backpropagation through time (TBPTT) [Mozier, 2013, Robinson and Fallside, 1987, Werbos, 1988]:
870 the forward pass unrolls the entire trajectory, but gradients are propagated only through the most
871 recent $w_{bptt}(e)$ steps at epoch e . Hidden states are passed forward between segments, while earlier
872 segments are treated as stop-gradient.

873 We employ a curriculum learning approach analogous to scheduled sampling. Let T_{max} be the
874 horizon (300 iterations). With a warm-up of e_w epochs,

$$w_{bptt}(e) = \begin{cases} w_{start} & e \leq e_w \\ \min\left(T_{max}, w_{start}\gamma_{bptt}^{\lfloor \frac{e-e_w}{f_{bptt}} \rfloor}\right) & e > e_w \end{cases} \quad (12)$$

875 so the window grows geometrically by a factor of $\gamma_{bptt} > 1$ every f_{bptt} epochs and is capped at the
876 full trajectory length.

877 Table 3 contains all hyperparameter details for the LSTM routers. The routers for the Paired Solver
878 experiment took a maximum of 4 hours and 30 minutes to train while the routers for the Solver
879 Ensemble experiment took a maximum of 6 hours to train. We then use the model with the best
880 validation loss for testing. The architecture and dataset size is larger for the Solver ensemble
881 experiments to encourage the model to learn some of the nuanced differences between the classes.

Table 3: Hyperparameter settings for routers in Paired solver and Solver Ensemble Experiments

Hyperparameter	Paired Solver	Solver Ensemble
Learning rate	1e-3	1e-3
Hidden dimension	256	512
No. of hidden layers	4	3
Gradient Clipping Norm	1.0	1.0
Weight Decay	0.005	0.005
Batch size	32	32
Training samples	256	1024
Validation samples	32	128
Epochs	200	200
ss_{start}	1.0	1.0
γ_{tf}	0.95	0.95
ss_{end}	0.0	0.0
w_{start}	50	50
γ_{bptt}	1.25	1.25
e_w	10	10
f_{bptt}	1	1

D Additional Experimental Results

The original submission’s Dirichlet experiment (31×31 grid, LSTM router, MLP-branch DeepONet), which this appendix previously kept, has been withdrawn. A review of its evaluation code found that HINTS’ predictions had been mean-centred before comparison with the (uncentred) Dirichlet reference solution while the other methods’ predictions had not, which penalises HINTS by an amount we cannot reconstruct without the original per-instance predictions; in addition, its symmetric-GS rows used the mis-scaled SSOR reference noted in Section E.1. The evaluation code has been corrected (`results.py`), but the experiment has not been rerun, so the wall-clock study of Section E, which is restricted to periodic boundary conditions, is the paper’s only experimental evidence. Extending the corrector to Dirichlet boundaries requires a sine-transform trunk in place of the Fourier trunk and is left to future work.

E Cost-Aware Routing and Wall-Clock Evaluation

The experiments of Section 7 compare methods per iteration, which favors methods whose iterations are expensive. This appendix evaluates the end-to-end *wall-clock time to a target accuracy* of the learned router against HINTS, the classical solver alone and the classical methods a practitioner would reach for on these problems (geometric multigrid and preconditioned Krylov methods), for all five classical solvers and three PDEs, on 128^2 , 256^2 and 512^2 grids, using the cost-aware instantiation of the greedy rule introduced at the end of Section 4. Besides the Poisson and convection–diffusion equations of Section 7 we add *variable-coefficient diffusion* (Section E.5), on which no Fourier diagonalisation exists and the corrector must be learned from data, and *anisotropic diffusion*, $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$ with $\epsilon = 10^{-2}$ (diffusion tensor $\text{diag}(\epsilon, 1)$), on the same periodic domain with the same forcing distribution. It is the standard model problem on which point-relaxation methods fail to smooth: an error component oscillatory in x_1 and smooth in x_2 is damped by only $(1 - \epsilon)/(1 + \epsilon) \approx 0.98$ per Jacobi sweep, and geometric multigrid with point smoothing loses its grid-independent convergence (line relaxation or semi-coarsening would be needed). Section E.1 describes the protocol; Sections E.2 to E.5 collect all results per equation (every grid, every tolerance, every baseline, with significance tests and seed trials); Section E.7 studies ensembles of one, two and three classical solvers; Section E.8 accounts for overheads and training cost; and Section E.10 verifies numerically that the assumptions of Theorem 4.1, Propositions 4.2–4.3 and Theorem 5.1 hold in these settings.

E.1 Protocol

Fast, matched implementations. A wall-clock comparison is meaningful only if the classical baseline is not handicapped by implementation. All classical operations are therefore implemented as compiled, single-threaded C kernels on the periodic 5-point stencil (`stencil.c`, built with `cc -O3`): the residual together with its norm (the stopping test, in one pass), fused (damped) Jacobi sweeps, in-place lexicographic Gauss–Seidel, SOR and symmetric SOR sweeps (forward and backward), a lexicographic line Gauss–Seidel sweep in which every grid line along the strongly coupled axis is solved exactly by a cyclic Thomas algorithm (the line relaxation that anisotropic problems call for), and the full-weighting and bilinear transfers of geometric multigrid, whose V(2,2)-cycles use Gauss–Seidel smoothing—or line Gauss–Seidel smoothing for the anisotropic equation—with rediscretised coarse operators and a direct solve on the 32×32 coarsest grid (error contraction ≈ 0.03 per cycle on the isotropic equations and, with line smoothing, on the anisotropic one; point-smoothed multigrid contracts the anisotropic error by only ≈ 0.8 per cycle). At 128^2 a residual evaluation with its norm costs **[pending]**, a Gauss–Seidel sweep **[pending]**, a V-cycle **[pending]** and one corrector call **[pending]**, all measured in the exact per-iteration form in which the benchmark charges them (Table 7); an earlier version of this study used numpy/scipy sweeps that were $4\text{--}5\times$ slower and correspondingly overstated the router’s margins (Section E.9). The compiled kernels agree with the numpy/scipy implementations they replace to 2×10^{-14} (relative), the line sweep agrees with a sparse block-triangular solve to 10^{-13} , and their operators and one-step iterates match the dense reference implementations of the original submission to the reference’s own precision (relative error below 10^{-7} , the reference assembling its matrices in single precision; 10^{-12} where the reference is exact); the comparison is an assertion in `validate_fast_pde.py`. In the course of this check we found that the original dense symmetric-GS reference mis-implemented the SSOR scaling (the factor

935 $\omega/(2 - \omega)$ was added to D^{-1} instead of multiplying it); it has been corrected, and the wall-clock
 936 study uses the standard SSOR throughout. Reference solutions are computed by FFT diagonalization
 937 of the same discrete operator, i.e., they are exact solutions of the discrete system, so true errors are
 938 available for evaluation at no charged cost; the FFT solve is also reported as a baseline in every table
 939 (it is a direct method that exists because these test problems have constant coefficients and periodic
 940 boundaries, and it is faster than every iterative method, including ours; the variable-coefficient
 941 problem of Section E.5 has no such solve). All timings are single-threaded CPU measurements
 942 (Apple M4 Pro, numpy/scipy/PyTorch, one thread). The benchmark guards against machine drift:
 943 before the timed replays of every test instance it re-times a reference operation (one corrector call)
 944 and waits, in 5 s steps, until that time is within 10% of its value at the start of the session; the largest
 945 deviation accepted in the study is [pending]%, the guard waited 0 times in total, and the per-instance
 946 ratios and load averages are stored with the results (an earlier confirmatory run had been contaminated
 947 by a 3.5 h stall of the machine inside one cell, which the harness could not detect). Two accounting
 948 checks are reported in Section E.8: the charged time of a router solve is [pending]–[pending]% of
 949 the outer elapsed time of its replay, and its live time to h^2 is [pending]–[pending] times its work-unit
 950 time.

951 **Corrector.** The neural corrector is a DeepONet applied to the residual $r = f_h - \mathcal{L}_h^a u_h^{(t)}$ and
 952 evaluated on a 64×64 sensor grid, i.e., on the coarser grid obtained by band-limited (FFT) restriction
 953 to the modes $|k_1|, |k_2| \leq 31$ at every grid size (for the anisotropic equation the sensor grid is $N \times N/4$,
 954 i.e., full resolution along x_1 , so that the band contains every mode the point smoothers cannot damp;
 955 the sensor grid is the only PDE-specific hyperparameter of the pipeline; the variable-coefficient
 956 equation uses the isotropic 64×64 grid); its output is prolonged back to the fine grid by exact
 957 trigonometric interpolation, so the correction is band-limited by construction and never injects error
 958 into the high-frequency band that the classical smoother owns (Figure 2). The trunk net is the fixed
 959 orthonormal Fourier basis of that band ($p = 3969$ basis functions, as in POD-DeepONet [Lu et al.,
 960 2022]), and the branch net is a linear layer on the 4096 sensor values; it is fit by least squares on
 961 32,000–64,000 exact (residual, error) pairs of the discrete operator drawn from a mixture of spectral
 962 shapes (forcing-like, residual-like, white and randomly tilted spectra), and reaches a validation
 963 relative error of $\approx 2 \times 10^{-6}$ on every mode of the band. The corrector is applied scale-equivariantly,
 964 $C_{\text{NO}}(r) = \|Rr\| \mathcal{G}_\theta(Rr/\|Rr\|)$, which preserves $C_{\text{NO}}(0) = 0$ (Proposition 4.2). Its deployed form
 965 is therefore a linear, Fourier-structured map—a least-squares approximation of the exact band-limited
 966 inverse $P(\mathcal{L}_h)_{\text{band}}^{-1}R$ of the discrete operator. Replacing it by that exact inverse changes nothing
 967 (development run): on the GS pairing at 128^2 the cost-aware oracle’s median work-unit time to 10^{-8}
 968 was 4.472 ms with either corrector and its decision sequence identical on 13 of 16 instances. The
 969 gains reported below are thus those of an accurate band-limited spectral correction applied at the
 970 right moments; the learned corrector stands in for a spectral inverse that, on these constant-coefficient
 971 periodic problems, could also be computed directly. Its cost is dominated by a 4096×4096 matrix–
 972 vector product and is therefore almost independent of N ([pending] at 128^2 , [pending] at 512^2 ,
 973 Table 31), i.e., about [pending] Jacobi or [pending] GS sweeps at 128^2 and [pending] Jacobi sweeps
 974 at 512^2 (Tables 7, 13 and 19). The same corrector is used by HINTS and by all routing policies.

975 **Cost-equalised macro-actions.** Per-iteration costs c_j are measured once per pairing and grid on the
 976 benchmark machine, in the exact form in which the timed replay charges them—the update given the
 977 current residual, then the residual of the new iterate and its norm, i.e. the stopping test—as the median
 978 over seven interleaved blocks of the block median of 60 repetitions (the minimum and the spread
 979 across blocks are stored with the costs). The unit of cost u is one corrector call; every operation is
 980 applied $m_j = \max\{1, \text{round}(u/c_j)\}$ times per decision, and these macro-actions are compared by
 981 their plain error, i.e. the deployed rule is Algorithm 1 on macro-actions; only an operation dearer than
 982 the unit ($c_j > u$, hence $m_j = 1$) is compared per unit of cost, $(\|e_j\|/\|e^{(t)}\|)^{u/c_j}$, and the surrogate
 983 weights $\tilde{c}_j$ use the same exponent. With the compiled kernels every point smoother is cheaper than the
 984 unit on every grid and a multigrid cycle is cheaper than the unit at 128^2 and 256^2 , so the per-unit-cost
 985 comparison is used only for multigrid at 512^2 and, in the granularity ablation of Section E.6, for the
 986 corrector itself; everywhere else the deployed rule is exactly Algorithm 1 on macro-actions, which is
 987 also the rule the exhaustive check of Section E.10 verifies. Tables 7, 13 and 19 list c_j , u and m_j for
 988 every cell; the equalisation is exact up to rounding ($m_j c_j / u$ lies within $0.5/m_j$ of 1). Ensembles
 989 reuse the cached pairwise costs of their members, so every member has the same macro-action size in
 990 the ensemble as in its pairwise run. The cost-aware oracle runs Algorithm 1 on these macro-actions
 991 with the true error; the *greedy oracle* rows run the cost-agnostic Algorithm 1 on single operations,

as in the main text. (An earlier version of this study applied the exponent $u/(m_j c_j)$ to every action, including the rounding error of cheap ones, which is not Algorithm 1; the deployed rule was changed so that the theory covers it.)

Lightweight router. The learned router selects macro-actions from $6 + K$ scalar features, all $O(1)$ given the residual norm that any stopping test computes: the log relative residual, its per-iteration change over the last macro-action and an exponential moving average thereof, the log epoch index, a one-hot encoding of the previous macro-action, and the log counts of corrector calls so far and of iterations since the last call. It is a two-hidden-layer MLP (64 units) evaluated in numpy, costing $\approx 5 \mu\text{s}$ per decision in a warm loop and [pending]–[pending] μs in situ, where every decision after a macro-action finds cold caches (measured inside the timed replays, Section E.8), independently of N . It is trained with the surrogate of Section 5.1, with the per-state weights $\tilde{c}_k$ normalised by $\sum_k \tilde{c}_k$ (which does not change the Bayes decision), on states visited by oracle rollouts with ε -exploration (probability 0.1), followed by two to three DAgger rounds [Ross et al., 2011] in which the current router’s own rollouts are relabelled by the oracle—the batch analogue of the scheduled sampling used in Section C. Training uses 32–128 instances per round (fewer on the larger grids) drawn from the same GRF distribution as the test set but with a different seed; the router of every pairing is trained in seconds to a few minutes. Ensemble routers, which must additionally learn to choose among classical members, use a larger imitation budget (256 rollouts at 128^2 and 128 at 256^2 , six DAgger rounds, hidden width 128, 300 epochs): with the pairwise recipe the router over {Jacobi, Jacobi (0.67)} imitated its oracle imperfectly at tight tolerances (convection–diffusion, 128^2 , $\varepsilon = 10^{-8}$: 3.34 against the oracle’s 2.67 work-unit ms), which the larger budget removes (2.70). At 512^2 the routers are trained from 32 oracle rollouts of at most 400 iterations (stopping error 10^{-8}); for the convection–diffusion Jacobi pairing, where the first such router missed h^2 within the cap on 4 of 16 instances, the reported router uses 64 rollouts of at most 800 iterations—the only per-cell change to the recipe.

Baselines and metric. HINTS applies the corrector every τ -th iteration, first at iteration τ (Section 3). Zhang et al. [2024] use the proportion $1/15$ (one DeepONet call per 14 relaxation sweeps) in all of their two- and three-dimensional experiments and $1/25$ only in the one-dimensional Poisson experiment; since all our problems are two-dimensional, the reference schedule is $\tau = 15$, and $\tau = 25$ is reported as well (an earlier version of this study used it as the reference; the corresponding margins are [pending]–[pending] at h^2 and [pending]–[pending] at 10^{-8} on the 128^2 pairwise cells). We further report the *best fixed schedule*, selected *per cell and per tolerance* on the test instances themselves (an optimistic baseline), from the family of HINTS with $\tau \in \{2, 5, 10, 15, 25, 50\}$, phase-shifted HINTS with the first call at iteration 0 and $\tau \in \{5, 10, 15, 25, 50\}$, and the *one-shot schedule*—the corrector once at iteration 0, then the classical solver only—which is the static rule closest to what the router learns at truncation level and therefore the natural ablation for state-dependent routing ($\tau = 1$, the corrector at every iteration, never reduces the error outside the corrector’s band and is omitted); and the best of three *residual-decay rules*, deterministic controls that decide at the router’s granularity from the same residual norm that the stopping test computes: the corrector at iteration 0, then the solver’s macro-action, and the corrector again whenever the per-iteration contraction of the residual over the last macro-action was worse than $\theta \in \{0.9, 0.95, 0.98\}$ (the threshold is selected per cell and tolerance on the test set, like the best fixed schedule). The classical methods without corrector run on the same stencil, with the same accounting, *inside the same replay loop as the policies*—same instance, same warm-up, same random order, same three replays: multigrid V(2,2) alone; conjugate gradients, CG preconditioned by symmetric GS and by a symmetric V-cycle (forward and backward GS, both compiled) for Poisson; BiCGSTAB, multigrid-preconditioned BiCGSTAB and restarted GMRES(20) for convection–diffusion (GMRES is accounted at the granularity of its restart cycles and is not run at 512^2); point-smoothed and line-smoothed multigrid, CG and its two preconditioned variants for anisotropic diffusion; and the sparse direct solve with a cached factorisation together with the multigrid and CG variants for variable-coefficient diffusion. The Krylov methods are SciPy implementations driven by the compiled matrix–vector product and preconditioners: their per-iteration true errors come from an untimed pass with a callback, and the timed pass runs exactly the same number of iterations with a timestamp-only callback (a few hundred nanoseconds per iteration) and verifies that its final iterate reaches the error the untimed pass recorded (largest ratio of the two final errors over the study: [pending]). An earlier version of this study timed the classical baselines in separate sessions; the same computation differed by 4–9% between sessions on identical instances (paired, $p < 10^{-4}$), so every cross-session p -value was

1049 uninformative, which is why all baselines are now timed in-session. All configuration choices of
 1050 the study were made on a development test set (GRF seed 72, with the earlier numpy kernels, the
 1051 earlier forcing sampler and $\tau = 25$ as reference; archived as `results_dev/`); the results reported
 1052 here are a single confirmatory evaluation of the frozen configurations on a fresh test set (seed 73)
 1053 with three timed replays per instance, and Section E.9 places the two side by side. The test forcings
 1054 are drawn from the hierarchical GRF of Section 7 as real white noise filtered in Fourier space, which
 1055 realises exactly the stated covariance with the Hermitian symmetry of a real field (the development
 1056 runs used a sampler that drew independent complex coefficients on the half-spectrum, which halved
 1057 the variance of the modes on the two coordinate axes); the Nyquist modes are excluded, so every
 1058 forcing is a trigonometric polynomial with $|k| < N/2$ in both directions, and the instances are drawn
 1059 sequentially so that the first n instances of any script are the benchmark’s. For each test instance (64
 1060 at 128^2 , 32 at 256^2 , 16 at 512^2) we record, after every iteration, the true relative error $\|e_h^{(t)}\|/\|u_h\|$
 1061 (untimed, benchmark-only) and the cumulative charged wall-clock time of a replay that executes
 1062 only the chosen operations (residual and its norm, update, and—for the learned router—every feature
 1063 and decision computation; oracle decisions are not charged). The reported time to a tolerance is
 1064 the charged time at the first iteration whose true error is below it, identically for every method: it
 1065 is a benchmark of the work needed to reach an error level, not of a deployed stopping rule (which
 1066 would stop on the residual norm that every method computes anyway). We report the median first
 1067 time at which the error drops below a tolerance ε ; paired speedups—the median over instances of
 1068 the per-instance time ratio, which is the estimand behind every speedup, bold mark and test in this
 1069 appendix (medians of raw times are shown for scale and can order differently from the paired median
 1070 on small samples)—with bootstrap 95% intervals (2000 resamples over instances) and two-sided
 1071 Wilcoxon signed-rank tests on the paired log time ratios (the direction is read from the median ratio;
 1072 “slower” marks a median below one); the study performs several hundred such tests, so the speedup
 1073 tables mark as significant only comparisons that survive a Holm correction at level 0.05 over all
 1074 tests of the table. A run that does not reach ε within the iteration cap (60,000 at 128^2 , 40,000 at
 1075 256^2 , 8,000 at 512^2 ; 5,000 units for the classical baselines, 3,000 at 512^2) is marked $\dagger$ with the
 1076 number of censored instances, reported as a lower bound when it is the majority, and enters every
 1077 paired ratio and test at the time it spent up to the cap, on whichever side it occurs (a router failure
 1078 therefore counts against the router; a ratio against a majority-censored baseline is marked $\geq$ as a
 1079 lower bound)—a labelled lower-bound analysis rather than a survival analysis. Timed replays are
 1080 repeated three times per instance in a random order over policies and baselines, each replay preceded
 1081 by an untimed warm-up (one corrector call and one sweep, or the baseline’s own operation) so that
 1082 every method starts from the same cache state—the corrector’s matrix exceeds the CPU caches,
 1083 and without the warm-up a policy’s first call was up to 0.1 ms slower or faster depending on which
 1084 policy had run before it; the median over replays is used. The measured per-operation costs set the
 1085 macro-action sizes and define *work units* (measured per-iteration costs times executed operations,
 1086 plus the measured decision cost per decision), a timer-free measure of the decision sequence used
 1087 wherever runs from different sessions must be compared; all reported live times are measured directly.
 1088 We highlight $\varepsilon = h^2$ as a nominal grid-dependent tolerance and verified on the confirmatory instances
 1089 themselves (**[pending]** instances over all equations and grids) that it lies below their discretisation
 1090 error: comparing the exact discrete solution with the exact discrete solution on a four times finer grid
 1091 for the same forcing (`discretization_error.py`; for variable coefficients the coefficient is one
 1092 grid-independent function), the relative discretisation error has median **[pending]–[pending]** h^2 over
 1093 the equations and grids and is at least **[pending]** h^2 on every instance, so driving the algebraic error
 1094 below h^2 is very unlikely to improve the computed solution materially (the two error components can
 1095 partly cancel, so “never” would be too strong); we also report 10^{-3} and 10^{-8} , and the iteration-based
 1096 metrics of the main text ($\text{AUC} = \sum_{t=1}^T \|e_h^{(t)}\|/\|u_h\|$ over $T = 300$ iterations, compared by paired
 1097 t -tests on log AUC) on the 128^2 grid. Every router was additionally retrained from scratch with five
 1098 independent seeds (different training instances, exploration and initialisation; the corrector and the
 1099 test set are fixed) and re-run on the test set; because those trials ran in separate sessions, whose live
 1100 timings are not comparable with the main run’s, they are compared in work units.

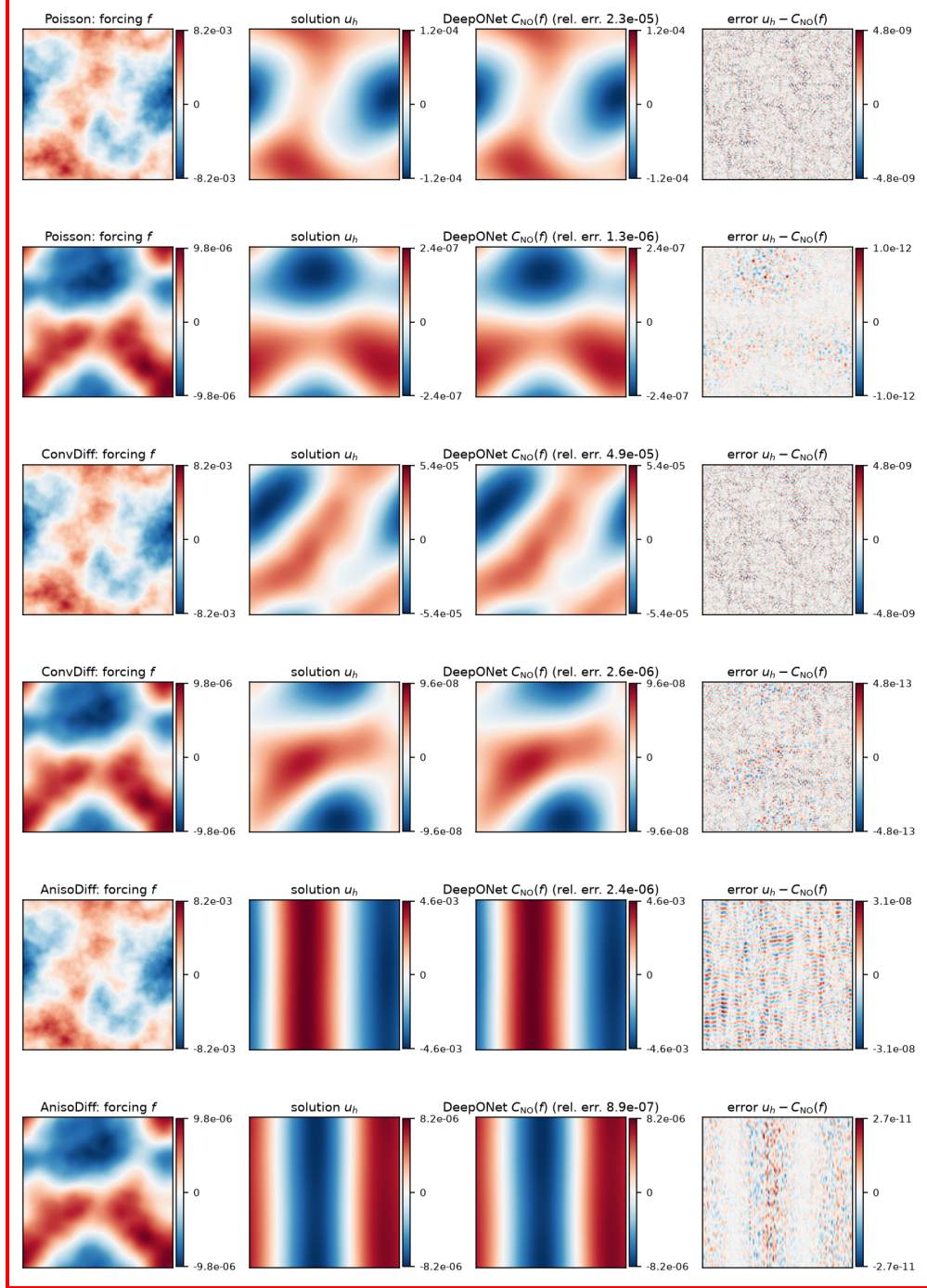


Figure 2: One-shot corrector predictions on the 128×128 grid (two test instances per equation): forcing, exact discrete solution, the DeepONet correction applied to the initial residual ($u_h^{(0)} = 0$), and the remaining error, which is confined to the modes outside the corrector's band (the relative error of the prediction is 10^{-6} – 10^{-5} on the isotropic equations and 10^{-6} on anisotropic diffusion, whose solution is dominated by modes that are smooth in x_2).

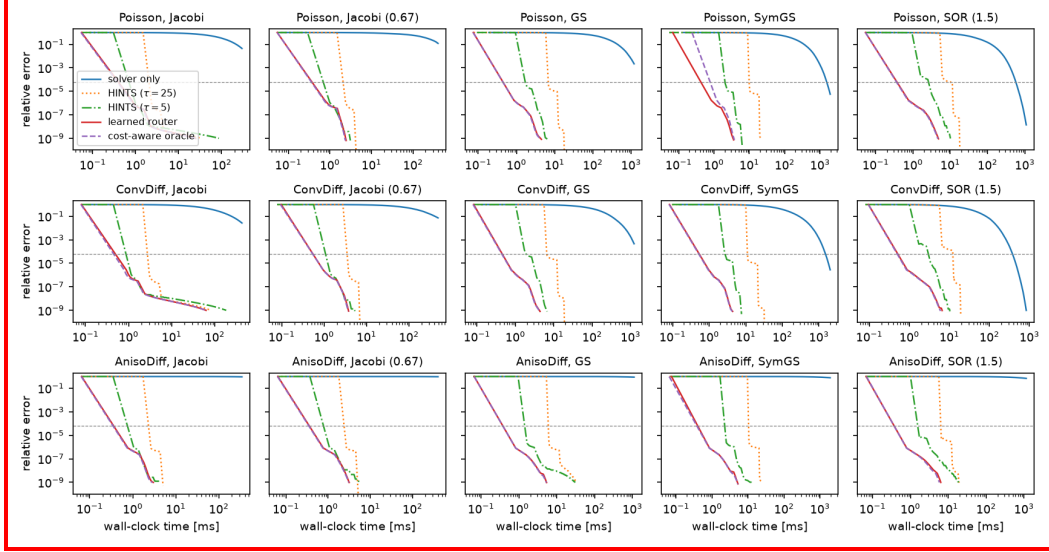


Figure 3: Relative error versus charged wall-clock time on the 128×128 grid for one representative test instance of every pairing (columns) and equation (rows); the dashed line is $\varepsilon = h^2$. The learned router and the cost-aware oracle are indistinguishable.

E.2 Poisson

Table 4 collects the median time to 10^{-3} , h^2 and 10^{-8} for every pairing, every policy and the classical baselines on all three grids; Table 5 the paired speedups of the learned router with their p -values; Figure 4 and table 6 where the iterations go; Table 7 the measured costs and macro-action sizes; Table 8 the iteration-based metrics; and Table 9 the seed trials.

Two regimes. At $\varepsilon = h^2$ the whole solve consists of one corrector call and a few sweeps (Table 6): the router, the cost-aware oracle, phase-shifted HINTS and the one-shot schedule take the same decisions and their times coincide up to timer resolution, so the router’s paired ratio against the best fixed schedule is close to one (Table 5; bold marks the comparisons that remain significant after the Holm correction), while HINTS with its published period, whose first call comes after $\tau - 1$ wasted sweeps, is several times slower, and the solver alone hundreds to thousands of times slower. Below h^2 the remaining error is the band the corrector cannot see, and the schedules separate: the router calls the corrector again only when the smooth error has re-emerged (two or three calls down to 10^{-8} ; Figure 4), the one-shot schedule never calls it again and pays for hundreds of extra sweeps, and every periodic schedule keeps paying for calls that no longer help, so at 10^{-8} the router beats HINTS and the one-shot schedule by large factors, whereas against the best fixed schedule and the best residual-decay rule—both selected on the test set—the outcome depends on the pairing (Table 5; the counts over all pairings are in Section 7, and Section E.6 traces the losses to the macro-action granularity). The cost-agnostic greedy oracle (Algorithm 1 on single operations) makes the same decisions as its cost-aware counterpart on the path to h^2 (they differ later, in when the corrector is called again); the learned router reproduces the cost-aware oracle’s decisions almost exactly (same median iteration and call counts, median times within a few percent), and the five retrained routers are compared with the main router in Table 9 [pending].

Classical baselines and larger grids. Against the classical methods without corrector (bottom block of Table 4; timed in the same session as the router of every pairing), the compiled multigrid is the strongest competitor: at 128^2 and $\varepsilon = h^2$ the stationary hybrids are [pending]–[pending] of a stand-alone V-cycle solve in paired ratio, and below h^2 they are slower than it ([pending]–[pending]), because one corrector call removes the modes $|k| \leq 31$ at the price of one matrix–vector product whereas a V(2,2)-cycle contracts the whole spectrum by ≈ 0.03 and keeps doing so; routing over {NO, multigrid} calls the corrector once and lets the cycles remove the rest ([pending]–[pending] of multigrid alone at h^2 and [pending]–[pending] at 10^{-8}), and against multigrid-preconditioned CG

the stationary hybrids are [pending]–[pending] at h^2 and [pending]–[pending] at 10^{-8} . The FFT direct solve, exact for this constant-coefficient periodic problem, is faster than every iterative method at every grid. At 256^2 and 512^2 [pending] the same two regimes are expected to hold with shrinking margins for the Jacobi and Gauss–Seidel pairings, because the corrector’s fixed band $|k| \leq 31$ leaves the smoother all modes $k \geq 32$, whose slowest member contracts by only 0.98 per Jacobi sweep at $N = 512$ (against 0.71 at $N = 128$): every schedule then needs the same ≈ 80 sweeps after the first call, the wall-clock time is dominated by sweeps, and the corrector—whose cost is nearly grid-independent—is worth fewer and fewer of them, so the best classical method at 512^2 is multigrid, alone or with one corrector call. This is the expected limit of a corrector whose band does not grow with the grid.

Table 4: Poisson: median wall-clock time to relative error $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$ on the 128^2 , 256^2 and 512^2 grids (64, 32 and 16 test instances) for every pairing $\{\text{NO}, \text{solver}\}$ and policy, and for the classical baselines without corrector. “Best fixed schedule” is the best of HINTS ($\tau \in \{2, 5, 10, 15, 25, 50\}$), phase-shifted HINTS ($\tau \in \{5, 10, 15, 25, 50\}$) and the one-shot schedule, and “Best residual-decay rule” the best threshold $\theta \in \{0.9, 0.95, 0.98\}$, both chosen per cell and tolerance on the test set itself (the chosen schedule is the subscript). The greedy oracle is the cost-agnostic Algorithm 1, the cost-aware oracle is Algorithm 1 on cost-equalised macro-actions; both use the true error and are not deployable. Bold marks the router where its median is the smallest among the deployable methods of its pairing (solver alone, every fixed schedule, the one-shot and residual-decay rules); the oracles are in italics; $\dagger n$: n instances did not reach the tolerance within the iteration cap ($>$: lower bound).

[results pending: confirmatory run in progress]

Table 5: Poisson: paired median speedup [bootstrap 95% interval] of the learned router at $\varepsilon = h^2$ and 10^{-8} over HINTS ($\tau=15$), over the best fixed period for that cell, over multigrid alone and over multigrid-preconditioned CG, with the one-sided Wilcoxon p -value that the router is faster (or, where its median is larger, that it is slower). Bold: $\geq 1.1 \times$ with $p < 0.01$.

[results pending: confirmatory run in progress]

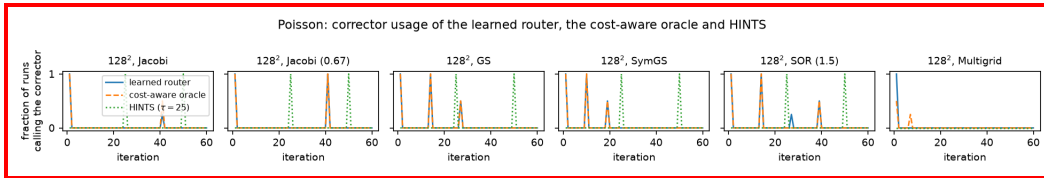


Figure 4: Poisson: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ($\tau=15$), for every pairing (columns) and grid (rows; all test instances of each grid).

Table 6: Poisson: median iterations (and corrector calls) to $\varepsilon = h^2$ for HINTS ($\tau=15$), the cost-aware oracle and the learned router on every grid.

[results pending: confirmatory run in progress]

Table 7: Poisson: measured cost of one classical iteration c_{solver} and of one corrector call c_{NO} (residual evaluation included), and the resulting macro-action size m_{solver} (sweeps per decision; the corrector is always applied once).

[results pending: confirmatory run in progress]

Table 8: Poisson, 128×128 : iteration-based metrics of the main text—relative error after $T = 300$ iterations and AUC over the T iterations (mean and standard error over 64 instances; p : one-sided paired t -test that the method’s AUC exceeds the router’s). Bold: router better than both solver-only and HINTS.

[results pending: confirmatory run in progress]

Table 9: Poisson, 128×128 : five independent router-training seeds per pairing—mean and standard deviation over seeds of the median work-unit time to h^2 , the share of test instances on which the seed router’s decision sequence to h^2 coincides with the main router’s, the range over seeds of the paired speedup over HINTS ($\tau=15$), and the number of seeds significantly faster ($p < 0.01$) than both HINTS ($\tau=15$) and the best fixed period.

[results pending: confirmatory run in progress]

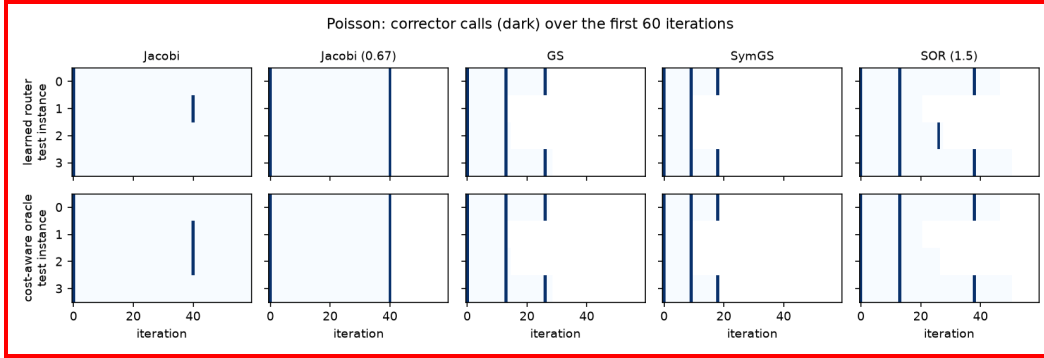


Figure 5: Poisson, 128×128 : routing decisions (dark: corrector call) of the learned router (top row) and the cost-aware oracle (bottom row) over the first 60 iterations of every test instance, for every pairing.

1142 E.3 Convection–diffusion

1143 The convection–diffusion results (Tables 10 to 15 and fig. 6) mirror the Poisson ones: at h^2 the router
 1144 is within a few percent of the best static schedule and $1.5\text{--}4\times$ faster than HINTS with the published
 1145 two-dimensional period, at 10^{-8} it beats HINTS and the one-shot schedule by large factors while
 1146 the comparison with the best fixed schedule and the best residual-decay rule is pairing-dependent
 1147 (Table 11), and it reproduces the cost-aware oracle; the nonsymmetric operator changes only the
 1148 details (the undamped-Jacobi pairing stalls at the near-Nyquist band below 10^{-6} for every policy,
 1149 and GMRES(20) is two orders of magnitude slower than the router).

Table 10: Convection–diffusion: median wall-clock time to relative error $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$ on the 128^2 , 256^2 and 512^2 grids for every pairing and policy, and for the classical baselines without corrector (conventions as in Table 4).

[results pending: confirmatory run in progress]

Table 11: Convection–diffusion: paired median speedup [bootstrap 95% interval] of the learned router at $\varepsilon = h^2$ and 10^{-8} over HINTS ($\tau=15$), the best fixed period, multigrid alone and multigrid-preconditioned BiCGSTAB, with two-sided Wilcoxon p -values (conventions as in Table 5).

[results pending: confirmatory run in progress]

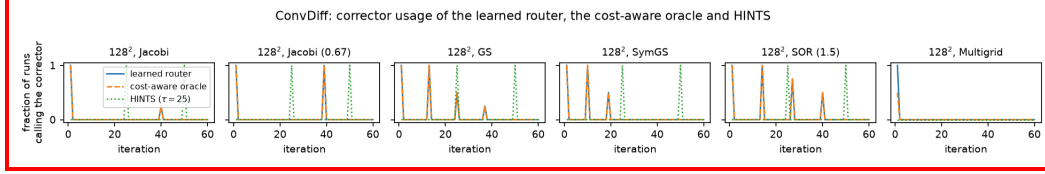


Figure 6: Convection–diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ($\tau=15$) for every pairing (columns) and grid (rows).

Table 12: Convection–diffusion: median iterations (and corrector calls) to $\varepsilon = h^2$ on every grid.

[results pending: confirmatory run in progress]

Table 13: Convection–diffusion: measured per-iteration costs and macro-action sizes.

[results pending: confirmatory run in progress]

Table 14: Convection–diffusion, 128×128 : iteration-based metrics (conventions as in Table 8).

[results pending: confirmatory run in progress]

Table 15: Convection–diffusion, 128×128 : five-seed router-training trials (conventions as in Table 9).

[results pending: confirmatory run in progress]

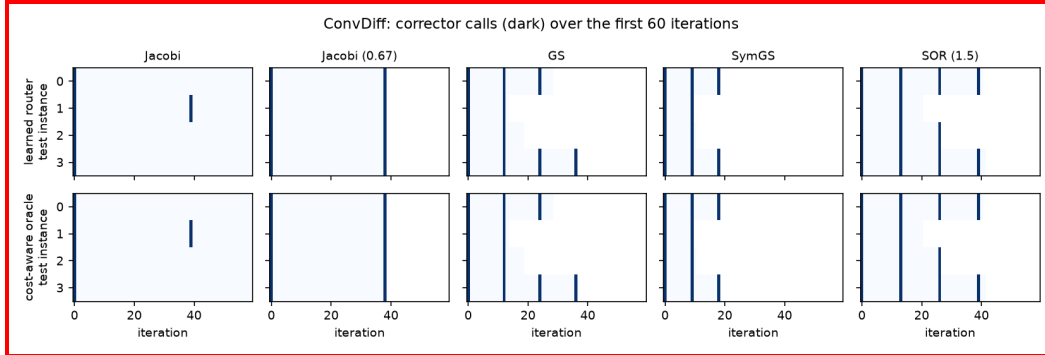


Figure 7: Convection–diffusion, 128×128 : routing decisions (dark: corrector call) of the learned router (top row) and the cost-aware oracle (bottom row) over the first 60 iterations of every test instance.

1150 E.4 Anisotropic diffusion

1151 The anisotropic equation is where the two components of the hybrid are most complementary and
 1152 where the point smoothers are weakest: they cannot damp the x_1 -oscillatory modes at all (Table 16:
 1153 none of the five point smoothers alone reaches h^2 within the iteration cap on either grid), whereas the
 1154 corrector, whose sensor grid keeps full resolution along x_1 , removes exactly those modes together
 1155 with the smooth ones in a single call and leaves the smoother only the x_2 -oscillatory band, which
 1156 it damps at the isotropic rate. The corrector’s $N \times N/4$ band is thus the spectral counterpart of
 1157 semi-coarsening, and the honest classical comparison is the method a practitioner would use on
 1158 this problem: *line* Gauss–Seidel along x_2 (which contracts the x_1 -oscillatory modes by ≈ 0.35
 1159 per sweep where point Jacobi contracts them by 0.98) and line-smoothed multigrid, which regains
 1160 grid-independent convergence (≈ 0.03 per cycle against ≈ 0.8 with point smoothing). Both are
 1161 included, line GS as the sixth pairing {NO, line GS} and, together with point-smoothed multigrid and
 1162 the CG variants, as classical baselines timed in the same session **[pending]**. For the point-smoother
 1163 pairings the router reaches h^2 in the time of one corrector call plus a few sweeps, as does the one-shot
 1164 schedule, while HINTS with the published period is 1.7–3.9 $\times$ slower (Table 17; first confirmatory
 1165 run, revision 1); the best fixed schedule, which is exactly one call at $t = 0$ followed by sweeps, is
 1166 3–7% faster than the router at h^2 , the cost of the router’s per-step decision. At 10^{-8} the router was
 1167 significantly faster than the best fixed schedule in two of the five point-smoother pairings (Jacobi and
 1168 damped Jacobi) and at parity in the other three, within 4–7% of the cost-aware oracle; these numbers
 1169 are being regenerated under the revised protocol **[pending]**, and the five-seed trials for this equation
 1170 are **[pending]** (Table 21). The 512^2 grid is not run: the anisotropic corrector keeps full resolution
 1171 along x_1 ($N \times N/4$ sensors), so at 512^2 its least-squares branch would be a dense $65,536 \times 65,536$
 1172 matrix (17 GB in single precision), which the current design cannot hold; a coarser x_1 resolution
 1173 would leave the point smoothers modes they cannot damp (Section E.7 discusses exactly this regime).

Table 16: Anisotropic diffusion: median wall-clock time to relative error $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$ on the 128^2 (64 instances, six pairings including line GS) and 256^2 (16 instances) grids for every policy, and for the classical baselines without corrector (conventions as in Table 4; the point-smoother-only runs are capped at 60,000 and 40,000 iterations and never reach h^2).

[results pending: confirmatory run in progress]

Table 17: Anisotropic diffusion: paired median speedup [bootstrap 95% interval] of the learned router at $\varepsilon = h^2$ and 10^{-8} over HINTS ($\tau=15$) and the best fixed period, with two-sided Wilcoxon p -values and bootstrap intervals; the classical baselines (point- and line-smoothed multigrid, CG variants) are timed in the same session.

[results pending: confirmatory run in progress]

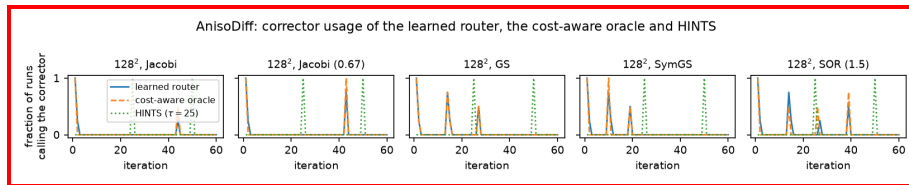


Figure 8: Anisotropic diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ($\tau=15$) for every pairing (columns) and grid (rows).

Table 18: Anisotropic diffusion: median iterations (and corrector calls) to $\varepsilon = h^2$.

[results pending: confirmatory run in progress]

Table 19: Anisotropic diffusion: measured per-iteration costs and macro-action sizes.

[results pending: confirmatory run in progress]

Table 20: Anisotropic diffusion, 128×128 : iteration-based metrics (conventions as in Table 8).

[results pending: confirmatory run in progress]

Table 21: Anisotropic diffusion, 128×128 : five-seed router-training trials (conventions as in Table 9).

[results pending: confirmatory run in progress]

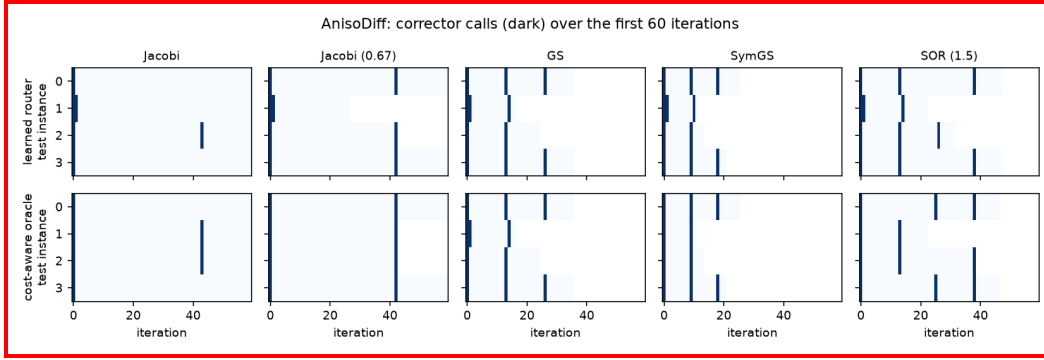


Figure 9: Anisotropic diffusion, 128×128 : routing decisions (dark: corrector call) of the learned router (top row) and the cost-aware oracle (bottom row) over the first 60 iterations of every test instance.

1174 E.5 Variable-coefficient diffusion

1175 The three equations above have constant coefficients on a periodic domain, so their discrete operators
 1176 are diagonalised by the FFT: the exact band-limited inverse that the corrector approximates has a
 1177 closed form, and a direct FFT solve is faster than any iterative method (Tables 4 and 10). To evaluate
 1178 the hybrid where neither exists we add $-\nabla \cdot (a(\mathbf{x})\nabla u) = f$ on the same periodic domain with a
 1179 fixed, smooth, positive coefficient field $a(\mathbf{x}) = \exp(\kappa z(\mathbf{x}))$, where z is a band-limited random field
 1180 ($|k| \leq 4$, one fixed realisation, scaled by its maximum on a fixed 1024^2 reference grid, so that a is
 1181 one grid-independent function sampled on every grid) and $\kappa = \frac{1}{2} \ln 10$, so that $\max a / \min a = 10$;
 1182 the forcing distribution is unchanged. The discretisation is the conservative 5-point scheme with face-
 1183 averaged coefficients (symmetric positive semidefinite, constants in its null space); reference solutions
 1184 come from a sparse LU factorisation with one pinned unknown, which is also reported as a direct-
 1185 solve baseline (its solve time with the factorisation cached, the factorisation itself costing ≈ 0.5 s at
 1186 128^2). Geometric multigrid rediscratises the coarse operators with block-averaged coefficients and
 1187 contracts the error by ≈ 0.1 per V(2,2)-cycle. The corrector is the same least-squares DeepONet
 1188 as before—a linear map on the 64×64 band—but fitted to (residual, error) pairs of *this* operator,
 1189 whose band-limited inverse $R(\mathcal{L}_h)^{-1}P$ is not diagonal in the Fourier basis and has no closed form.
 1190 It is nevertheless a fixed 4096×3969 matrix that could be assembled directly from 4096 sparse
 1191 solves, one per sensor basis function; the least-squares fit on random pairs is a cheaper approximation
 1192 of the same matrix, with validation error on the band of order 10^{-2} rather than 10^{-6} . What this
 1193 experiment tests is therefore routing with an inexact corrector on an operator without Fourier structure
 1194 (no shared eigenbasis with the smoothers), not that the correction must be learned: because the
 1195 corrector is inexact it is called repeatedly along a solve, and the router’s decisions matter beyond
 1196 the first call. The coefficient field is fixed rather than sampled, so this experiment tests a corrector
 1197 learned for one operator without an analytic inverse, not generalisation across coefficient fields; the
 1198 latter would require the coefficient-conditioned branch of the original DeepONet and is left to future

1199 work. Tables 22 to 27 and fig. 10 report the same quantities as for the other equations on the 128^2
 1200 and 256^2 grids.

Table 22: Variable-coefficient diffusion: median wall-clock time to relative error $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$ on the 128^2 and 256^2 grids for every pairing and policy, and for the classical baselines without corrector, including the sparse direct solve (conventions as in Table 4).

[results pending: confirmatory run in progress]

Table 23: Variable-coefficient diffusion: paired median speedup [bootstrap 95% interval] of the learned router at $\varepsilon = h^2$ and 10^{-8} over HINTS ($\tau=15$), the best fixed schedule, the one-shot schedule, multigrid alone and multigrid-preconditioned CG, with Holm-corrected one-sided Wilcoxon significance (conventions as in Table 5).

[results pending: confirmatory run in progress]

(figure pending)

Figure 10: Variable-coefficient diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ($\tau=15$) for every pairing (columns) and grid (rows).

Table 24: Variable-coefficient diffusion: median iterations (and corrector calls) to $\varepsilon = h^2$.

[results pending: confirmatory run in progress]

Table 25: Variable-coefficient diffusion: measured per-iteration costs and macro-action sizes.

[results pending: confirmatory run in progress]

Table 26: Variable-coefficient diffusion, 128×128 : iteration-based metrics (conventions as in Table 8).

[results pending: confirmatory run in progress]

Table 27: Variable-coefficient diffusion: five-seed router-training trials (conventions as in Table 9).

[results pending: confirmatory run in progress]

1201 E.6 Decision granularity

1202 The cost-aware router decides once per macro-action, and the macro-action of a cheap solver is as
 1203 expensive as one corrector call—[pending] Jacobi or [pending] Gauss–Seidel sweeps at 128^2 with
 1204 the compiled kernels. Below truncation level the best fixed schedule is a phase-shifted HINTS that
 1205 calls the corrector every 5–10 sweeps, a schedule the cost-aware oracle cannot express because its
 1206 decisions are too coarse, which is why the router loses to it in several pairings at 10^{-8} (Tables 5, 11
 1207 and 17). The granularity is a free parameter of the construction of Section 4: the unit of cost can be
 1208 any fraction of a corrector call, actions dearer than the unit (here the corrector itself) being compared
 1209 per unit of cost. Table 28 repeats the 128^2 study with the unit set to one quarter of a corrector call
 1210 (about a quarter of those sweeps per decision; the corrector is then the one action dearer than the
 1211 unit and is scored per unit of cost), routers retrained with the same recipe, on the same confirmatory
 1212 instances, for every pairing and for the nested two-member Jacobi ensemble.

Table 28: Decision granularity at 128×128 : median time to h^2 and 10^{-8} of the cost-aware oracle and the learned router with the unit of cost equal to one corrector call (main results) and to one quarter of it, and the router’s paired speedup over the best fixed schedule at 10^{-8} in both cases.

[results pending: confirmatory run in progress]

1213 E.7 Solver ensembles

1214 **When can a second classical solver help?** The cost-aware oracle is a per-decision comparison
 1215 of macro-actions, so a router over $\text{NO} \cup \mathcal{W}$ can only beat the best pairwise router of its members
 1216 if, along the way to the tolerance, the error passes through regimes in which *different* members
 1217 give the largest reduction per unit cost. We screened this ceiling exhaustively before training any
 1218 router: for every subset $\mathcal{W}$ of up to three members of {Jacobi, Jacobi (0.67), GS, SymGS, SOR
 1219 (1.5), multigrid} (multigrid on the isotropic equations only), every equation and grid, we ran the
 1220 cost-aware oracle over $\text{NO} \cup \mathcal{W}$ on the test instances and compared its work-unit time to that of
 1221 the oracle over each member alone (Table 29). Two facts emerge. At $\varepsilon = h^2$ no ensemble beats
 1222 the best single solver anywhere: one corrector call followed by one or two macro-actions of the
 1223 cheapest smoother already reaches truncation-level accuracy, so there is no second regime to exploit.
 1224 At $\varepsilon = 10^{-8}$, after the corrector has removed the smooth band, the smoother has to damp the whole
 1225 band $|k| > 31$, whose sub-bands favour different solvers: undamped Jacobi is the cheapest per unit
 1226 cost on the modes near the band edge (it contracts them by $\cos(2\pi \cdot 32/N)$ -like factors that damped
 1227 Jacobi cannot match), but it does not damp the near-Nyquist modes at all, which damped Jacobi and
 1228 GS remove in a few sweeps. In the development screening ([provisional]: Table 29 is computed
 1229 with the development-run macro-action sizes; the confirmatory screening with the compiled kernels’
 1230 sizes is pending) the oracle over {Jacobi, Jacobi (0.67)}—a *nonstationary* Jacobi iteration whose
 1231 damping parameter is chosen per macro-action—was therefore faster than every single solver, by
 1232 13–28% on four of the six (equation, grid) settings at 128^2 and 256^2 , and adding a third member (GS)
 1233 changed nothing, i.e., the gain saturates. Whether this ceiling survives the coarser macro-actions of
 1234 the compiled kernels is exactly what the nested-ensemble runs below measure. At 512^2 multigrid
 1235 dominates every subset (its cycle removes all sub-bands at once), so the oracle over any ensemble
 1236 containing it equals the oracle over {NO, multigrid}: adding members neither helps nor hurts. The
 1237 “difficulty” that matters for ensembles is thus not the equation but the width of the spectrum left to the
 1238 classical members after the corrector, which grows with the grid (fixed band) and with the tolerance.

1239 **Nested ensembles.** Table 30 evaluates the learned router over the nested sets {Jacobi},
 1240 {Jacobi (0.67)}, {Jacobi, Jacobi (0.67)} and {Jacobi, Jacobi (0.67), GS} on every equation at 128^2
 1241 and 256^2 , and at 128^2 also over the four ensembles of Section 7 ({Jacobi, GS} up to all five solvers);
 1242 the single-member routers are the pairwise routers of Sections E.2 to E.4, evaluated in the same
 1243 session on the same instances, and all members keep the macro-action sizes of their pairwise runs.
 1244 The table reports live and work-unit times to h^2 , 10^{-6} and 10^{-8} for the router and its oracle, and
 1245 the paired speedup at 10^{-8} over the best single-member router (and of the ensemble oracle over
 1246 the best single-member oracle) with its p -value; work units are the primary measure, since they are
 1247 determined by the decision sequence alone. The 256^2 cells are [pending]; at 128^2 three things are
 1248 visible. *No separation at 128^2 :* at $\varepsilon = 10^{-8}$ the ensemble router is significantly faster than every
 1249 single-member router in [pending] of the [pending] ensemble cells, and the ensemble *oracle* is
 1250 [pending]–[pending] of the best single-member oracle in work units, i.e., the greedy rule itself finds
 1251 no second regime to exploit. This differs from the development runs, in which the two-member Jacobi
 1252 set was 8–15% faster than damped Jacobi alone on convection–diffusion and anisotropic diffusion
 1253 (Table 33), and the difference is the decision granularity. With the numpy sweeps a Jacobi macro-
 1254 action was about ten sweeps, and the oracle alternated undamped Jacobi—the cheapest member on
 1255 the band-edge modes $|k| \approx 32$, which it contracts by about 0.5 per sweep at 128^2 but which leaves
 1256 the Nyquist modes untouched—with damped Jacobi, which removes them. With the compiled sweeps
 1257 a macro-action is about forty sweeps: a single damped-Jacobi macro-action contracts the band-edge
 1258 modes by $0.665^{39} \approx 10^{-7}$ and the Nyquist modes by 0.34^{39} , so the whole band the corrector leaves
 1259 is gone after one or two macro-actions and there is nothing left for a nonstationary schedule to gain.
 1260 Section E.6 repeats these cells with a four times finer unit and tests directly whether the gain returns
 1261 [pending]; the 256^2 grids, where the band left to the smoothers is twice as wide and needs several
 1262 macro-actions, are the other place it could. *Saturation:* adding GS to the Jacobi pair changes the

1263 oracle’s schedule by at most a few percent (the three-member oracles match the two-member ones
 1264 within 1–2% in work units), because GS never wins a cost-aware comparison against the Jacobi pair.
 1265 *Monotonicity*: at h^2 every ensemble router is within [pending]–[pending] of the best single-member
 1266 router in work units, so enlarging the set is free at truncation level; at 10^{-8} the ensemble routers are
 1267 [pending]–[pending] of it, and [pending] of the [pending] cells are more than 5% slower, all of
 1268 them sets containing both GS and SymGS on convection–diffusion or anisotropic diffusion, where
 1269 the router’s imitation of its oracle degrades among near-tied members (the corresponding oracles are
 1270 within 6% of the best single oracle). A greedy rule is not monotone in its action set in general, and its
 1271 learned imitation is less so; in our settings the losses vanish at truncation level and reach [pending] at
 1272 10^{-8} .

Table 29: Oracle-level screening of ensembles ([provisional: development-run costs and instances]): for every equation and grid, the fastest single solver and the fastest ensemble of up to three members under the cost-aware oracle (median work-unit time over 16 test instances), at $\varepsilon = h^2$ and 10^{-8} ; the ratio is the speedup of the ensemble over the best single solver, and the last column is the ratio for {Jacobi, Jacobi (0.67)}.

[results pending: confirmatory run in progress]

Table 30: Nested ensembles: median time to $\varepsilon \in \{h^2, 10^{-6}, 10^{-8}\}$ of the learned router (live and work units) and of the cost-aware oracle (work units) over $\text{NO} \cup \mathcal{W}$, for the single members and the nested two- and three-member sets, all evaluated in one session on the same instances (64 at 128^2 , 32 at 256^2 , 16 for anisotropic diffusion at 256^2). The last columns give the paired speedup at 10^{-8} over the best single-member router (resp. oracle) with the one-sided Wilcoxon p -value; bold: $\geq 1.1 \times$ with $p < 0.01$.

[results pending: confirmatory run in progress]

1273 E.8 Overheads and amortisation

1274 Table 31 lists the measured cost of every operation as a function of N , including one decision of the
 1275 LSTM router of Section 7 (full-state input of size $4N^2$, hidden size 256, 4 layers): at 128^2 one LSTM
 1276 decision costs [pending] ms, about [pending] Jacobi sweeps, which is why the wall-clock study uses
 1277 the scalar-feature router ($\approx 5 \mu\text{s}$, independent of N). In situ, inside the timed replays, the router’s
 1278 decisions cost [pending]–[pending] μs each (the first decision of a solve and every decision after a
 1279 macro-action find cold caches; the $5 \mu\text{s}$ figure is a warm-loop median), i.e. [pending]–[pending]%
 1280 of the router’s time to h^2 ([pending]–[pending] decisions) and [pending]–[pending]% of its time to
 1281 10^{-8} ; the paired difference between the router and the cost-aware oracle at h^2 , whose decisions are
 1282 identical and uncharged, is [pending]–[pending] μs per solve and is exactly this decision cost (an
 1283 earlier version of this appendix put the decisions at “well under 1%” on the basis of the warm-loop
 1284 figure). The corrector calls account for 50–70% of the time to h^2 (one or two calls). Two further
 1285 accounting checks: the charged time of a router solve is [pending]–[pending]% of the outer elapsed
 1286 time of its replay, the remainder being the Python driver’s loop and timer overhead, which every
 1287 method pays per iteration; and the router’s live time to h^2 is [pending]–[pending] times its work-unit
 1288 time, the per-operation costs being measured in the same form as the replay charges them. Table 32
 1289 accounts for training: the corrector (shared with HINTS) is fit in minutes, the router in seconds to a
 1290 few minutes per pairing, and the break-even number of solves after which the router’s own training
 1291 cost is recovered relative to HINTS ($\tau=15$) with the same corrector is in the thousands at 128^2 and
 1292 in the hundreds on the larger grids.

Table 31: Measured single-thread cost of one operation as a function of the grid size (Poisson; median of repeated calls). Multigrid uses a direct solve below 64^2 and is not listed there.

[results pending: confirmatory run in progress]

Table 32: Training-cost amortisation for the GS pairing: corrector training (data generation and least-squares fit, CPU), router training (oracle rollouts and DAgger rounds), median wall-clock saved per solve at $\varepsilon = h^2$ relative to HINTS ($\tau=15$) and to the solver alone, and the number of solves after which the router’s training cost is recovered relative to HINTS.

[results pending: confirmatory run in progress]

1293 E.9 Development versus confirmatory runs

1294 Several choices of this study were made after looking at test results: the corrector band and the
 1295 anisotropic sensor grid, the router of the convection–diffusion Jacobi pairing at 512^2 (retrained with
 1296 more rollouts after the first router missed h^2 on 4 of 16 development instances, Section E.1), and
 1297 the imitation budget of the ensemble routers (raised after the pairwise recipe imitated its oracle
 1298 imperfectly on one development cell, Section E.7); the optimistic best-schedule baseline is tuned on
 1299 the test set by design and labelled as such. All of these were made on the development set (seed 72),
 1300 with the earlier numpy/scipy sweeps, the earlier forcing sampler and $\tau = 25$ as the HINTS reference
 1301 (Section E.1), which is why the table compares against $\tau = 25$ in both runs. Every result in this
 1302 appendix comes from a single evaluation of the frozen configurations on the confirmatory set (seed
 1303 73) with the compiled kernels and three timed replays. Table 33 lists, for every cell present in both,
 1304 the development and the confirmatory speedups: the two differ both through the instances and through
 1305 the kernels—the compiled sweeps are $4\text{--}5\times$ faster, so the corrector is worth $2\text{--}3\times$ more sweeps than
 1306 in the development runs and the margins over fixed schedules at h^2 shrink accordingly—and the table
 1307 shows exactly how much of the development-phase margin survives.

Table 33: Development (seed 72, numpy/scipy sweeps, one timed replay) versus confirmatory (seed 73, compiled kernels, three timed replays) paired median speedups of the learned router, for every pairwise cell at $\varepsilon = h^2$ against HINTS ($\tau=25$, the reference of the development runs) and against the best fixed schedule available in that run, and for every nested ensemble at 10^{-8} in work units against the best single-member router and against its oracle; two-sided Wilcoxon p -values in parentheses.

[results pending: confirmatory run in progress]

1308 E.10 Verification of the theoretical assumptions

1309 Theorem 4.1 and Propositions 4.2–4.3 assume that every error propagation map $G_j = I - C_j \circ \mathcal{L}_h^a$ is
 1310 Lipschitz with constant $\rho_j \leq 1$ (with at least one strict contraction in the optimal sequence, respec-
 1311 tively a shared unitary eigenbasis), zero-preserving, and—for postfix monotonicity—simultaneously
 1312 diagonalisable by a unitary matrix; Theorem 5.1 assumes that at every state at least one solver leaves
 1313 a positive fraction of the error. Table 34 checks these conditions numerically for every operation
 1314 of the study—finite diagnostics (a power-iteration estimate of the Lipschitz constant, commutators
 1315 on random vectors, exhaustive search over short horizons), none of which is a certificate over the
 1316 whole input space— on the mean-free subspace (the constant mode is the null space of the pe-
 1317 riodic operator and is excluded from the data and from every iteration): the Euclidean Lipschitz
 1318 constant $\|G_j\|_2$ (power iteration on $G_j^\top G_j$; exact from the Fourier symbol for Jacobi), the Lipschitz
 1319 constant $\|G_j\|_A = \|A_s^{1/2} G_j A_s^{-1/2}\|_2$ in the energy norm of the symmetric part A_s of the discrete
 1320 operator, the same for the macro-action $G_j^{m_j}$, the spectral radius, the smallest observed contraction
 1321 $\sigma_{\min} = \min \|G_j x\|/\|x\|$, $\|C_j(0)\|$, and the commutator $\|G_j G_{\text{NO}} - G_{\text{NO}} G_j\|$ on random vectors.
 1322 Table 36 evaluates the quantities of Theorem 4.1 and Theorem 5.1 along the cost-aware oracle’s own
 1323 path to $\varepsilon = 10^{-8}$ (a horizon of tens of macro-actions; to h^2 the path has one or two steps).

1324 **What holds, and in which norm.** (i) Every classical solver is zero-preserving exactly and contracts
 1325 in the sense that matters for convergence: the spectral radius of G_j is below 1 in every setting (at
 1326 most [pending]), and so is the energy-norm Lipschitz constant of the solver and of its macro-action
 1327 (at most [pending]). The *Euclidean* Lipschitz constant, however, exceeds 1 for Gauss–Seidel, SOR
 1328 and symmetric GS (up to [pending] for SOR), whose iteration matrices are non-normal; the proof of

Proposition 4.2 uses only the triangle inequality and the composition rule for Lipschitz constants, so it holds verbatim with $\|\cdot\|_2$ replaced by the energy norm, in which the condition is satisfied by all classical solvers in all settings—but that statement concerns the greedy rule and objective defined with the energy norm, whereas the oracle and the router of this study minimise the Euclidean error. For the evaluated Euclidean rule, the strict-contraction premise of Proposition 4.2 therefore holds for the Jacobi-type solvers and fails for the non-normal sweeps, whose Euclidean expansion is confined to a single sweep (their spectral radii and macro-action contraction show the iteration is convergent). For (damped) Jacobi both constants agree (the maps are Fourier-diagonal); the values are close to 1 because the slowest modes (the smoothest ones, and the near-Nyquist mode for undamped Jacobi) are barely damped by a point smoother—which is exactly why the corrector is needed. (ii) The corrector’s map is zero-preserving, contracts every band mode to **[pending]** or better (on average to 10^{-4} – 10^{-6}) and leaves the modes outside the band untouched, so its Lipschitz constant is exactly 1 (not < 1): Theorem 4.2 in its strict form does not apply to the corrector, but Theorem 4.3 does, and its second condition—a shared unitary eigenbasis—holds exactly for the exact band-limited inverse together with (damped) Jacobi, all of which are diagonal in the Fourier basis, and holds for the learned corrector up to its fit error (measured commutators $\approx 10^{-6}$, at the corrector’s single-precision accuracy; replacing the learned corrector by the exact inverse does not change the oracle’s decisions, Section E.1), so for the ensembles $\{\text{NO}, \text{Jacobi}\}$, $\{\text{NO}, \text{Jacobi} (0.67)\}$ and $\{\text{NO}, \text{Jacobi}, \text{Jacobi} (0.67)\}$ the objective is supermodular and postfix monotone up to that error and Theorem 4.1 applies with $\alpha = 1$; for GS, SOR, symmetric GS and multigrid the commutators are small but nonzero (10^{-3} – 10^{-1}) and no guarantee is claimed. (iii) Invertibility, which an earlier version of the theory used for postfix monotonicity, is neither needed for the corrected result nor sufficient in general (Section A.3); for the record, a Gauss–Seidel or symmetric-GS sweep annihilates an error concentrated at the first grid point ($\sigma_{\min} = 0$), SOR leaves $(1 - \omega)$ of it ($\sigma_{\min} = 0.72$ – 0.83 in Table 34), undamped Jacobi annihilates the mode $(N/4, N/4)$, and the corrector annihilates the band. Postfix monotonicity ($\mu = 1$) is therefore guaranteed only for the Jacobi-type ensembles, and is measured for the others. (iv) Table 35 checks Theorem 4.1 by exhaustive enumeration (in floating point, with errors clipped at a relative floor of 10^{-11}) for short horizons on the evaluated ensembles, in the Euclidean objective the oracle optimises and for the deployed cost-aware rule: for every pairing, for $\{\text{Jacobi}, \text{Jacobi} (0.67)\}$, for the two non-commuting sweeps $\{\text{GS}, \text{SOR}\}$, for $\{\text{GS}, \text{multigrid}\}$ and for all five point smoothers, the optimal sequence O of T macro-actions ($T = 4$, or 3 for $K = 6$) is found by exhaustive search on 8 test instances (4 for $K = 6$), and μ (over the greedy prefixes), $\alpha(O)$ (over the greedy prefixes, and over every sequence of length below T) and the bound itself are computed. In every ensemble $\mu \leq \text{[pending]}$ and $\alpha(O) = 1$ along the greedy prefixes; the bound holds on all **[pending]** instance–ensemble pairs (on **[pending]** of them the exhaustive optimum is at the round-off floor, where the bound holds by construction); the deployed rule differs from Algorithm 1 on macro-actions on **[pending]** of them; no prefix violates the weak-supermodularity inequality with a nonpositive denominator (**[pending]** such prefixes); and the greedy sequence coincides with the exact optimum or is within a few percent of it on most instances (the exceptions are myopic misses of up to **[pending]**× on single instances, most often for $\{\text{GS}, \text{SOR}\}$ on convection–diffusion); over all short prefixes the ratio exceeds 1 only for the SOR pairing, and only on prefixes the greedy rule never visits. The check supports the optimality of the greedy rule over a few macro-actions; the horizons at which the oracle runs to 10^{-8} are 10–400 macro-actions (Table 36), where the only available constant is the worst-case bound of Proposition 4.2. The worst-case $\alpha(O)$ of Theorem 4.2, which only uses the Lipschitz constants, is loose in comparison: along the oracle’s path to 10^{-8} (Table 36) $\sum_i \rho_{O_i}^2$ is close to T because each ρ is dominated by the slowest mode, so the bound ranges from **[pending]** (the multigrid pairing, whose cycle contracts everything) to 10^2 – 10^4 for the point smoothers, and is infinite on the few instances where $\sum_i \rho_{O_i}^2 \geq T$. (The last two columns of Table 36 evaluate the ratio of Equation (7) with the oracle’s own suffix in place of the unknown long-horizon O ; this is only a diagnostic, and its maximum is attained trivially at the last prefix.) (v) Theorem 5.1 is stated for the per-state normalised weights $\tilde{c}_j / \sum_k \tilde{c}_k$ that the router is trained with; its one hypothesis is that at every visited state at least one solver leaves a positive fraction of the current error, i.e. $\max_j \tilde{c}_j / \|e^{(t)}\|^2 > 0$ (last column of Table 36; the column before it shows that the per-step errors never exceed the current error), so the surrogate’s Bayes consistency applies at every visited state.

Table 34: Numerical check of the assumptions of Propositions 4.2–4.3 for every operation of the study (mean-free subspace). **[Provisional: development run; the confirmatory run regenerates this table with the compiled kernels’ macro-action sizes.]** $\|\cdot\|_2$: Euclidean Lipschitz constant of the error propagation map (power-iteration estimate; exact from the symbol for Jacobi); $\|\cdot\|_A$: Lipschitz constant in the energy norm of the symmetric part of $\mathcal{L}_h$, for one iteration and for the macro-action of m_j iterations; ρ : spectral radius; $\sigma_{\min}$: smallest observed contraction (a unit error at the first grid point for the triangular sweeps, the minimum over Fourier modes otherwise; for the corrector, the largest remaining fraction of a band mode); $\|C_j(0)\|$: zero preservation; last column: commutator with the corrector’s map on random unit vectors (Proposition 4.3). For (damped) Jacobi the constants are exact from the Fourier symbol (the value with the Nyquist modes included; without them undamped Jacobi has $\|G\|_2 = \rho = 0.9994$ at 128^2).

[results pending: confirmatory run in progress]

Table 35: Exhaustive short-horizon check of Theorem 4.1 (128×128 , n test instances per row, T macro-actions, K actions including the corrector, deployed cost-aware rule): the optimal sequence O is found by exhaustive search; $\mu = \max_{t < T} g(S^t \oplus O)/g(O)$ over the greedy prefixes; $\alpha(O)$ over the greedy prefixes and over all sequences of length below T ; whether $g(S^T) \leq \mu(1 - \phi_T)g(O) + \phi_T g(\emptyset)$ holds; the ratio $g(S^T)/g(O)$ of the greedy to the optimal error (median and maximum over instances); the number of instances on which the greedy sequence is the optimum; and the numbers of instances on which the exhaustive optimum is at the round-off floor (bound trivially true), on which the deployed rule coincides with Algorithm 1, and of prefixes at which the weak-supermodularity inequality fails with a nonpositive denominator. Errors below 10^{-11} relative are clipped to that floor.

[results pending: confirmatory run in progress]

Table 36: Quantities of Theorem 4.1, Proposition 4.2 and Theorem 5.1 along the cost-aware oracle’s path to $\varepsilon = 10^{-8}$ (8 test instances; medians unless stated): number of macro-actions T , $\sum_i \rho_{O_i}^2$ with the energy-norm constants of the chosen macro-actions, the per-instance margin $T - \sum_i \rho_{O_i}^2$ (median) and the resulting per-instance $\alpha(O)$ of Proposition 4.2 (median and range over the instances on which the margin is positive; ∞ otherwise), the empirical supermodularity ratio $\hat{\alpha}$ of Equation (7) evaluated with S the oracle’s prefix and S' its suffix (maximum and median over prefixes and instances), and the normalised bounds of Theorem 5.1: the largest ratio $\max_j \tilde{c}_j / \|e^{(t)}\|^2$ over visited states ($\bar{E}$) and the smallest value of $\max_j \tilde{c}_j / \|e^{(t)}\|^2$ ($E_{\min}$).

[results pending: confirmatory run in progress]

1384 F Limitations

1385 A primary limitation of our approach is the higher on-time computational cost requires to learn the
 1386 greedy routing strategy. In contrast, the simpler baselines such as HINTS incur no training overhead.
 1387 While this cost is amortized during inference time and leads to improved solution quality, it may be
 1388 prohibitive in settings with limited computational resources or when rapid deployment is required.
 1389 On the deployment side, Section E shows that the gains survive cost accounting: with a corrector
 1390 that covers the smooth half of the spectrum and a classical smoother that covers the rest, one or
 1391 two corrector calls suffice at truncation level, where the router reproduces the best static schedule
 1392 without a search; below truncation level its advantage over the best fixed schedule is not uniform
 1393 across pairings, and a compiled multigrid alone is faster there. The corrector we use is specific to
 1394 the constant-coefficient periodic setting (a fixed Fourier trunk basis on a coarse sensor grid), its cost
 1395 grows with the fourth power of the sensor-grid size, and undamped Jacobi remains limited far below
 1396 truncation level by the near-Nyquist modes that neither component damps. Our wall-clock study is
 1397 restricted to single-thread CPU execution; with a corrector band that does not grow with the grid,

1398 the gains over fixed schedules shrink at 512^2 for the point smoothers, whereas pairing the corrector
1399 with geometric multigrid is expected to restore them (Section E; [512² pending]). Ensembles of
1400 classical solvers pay off only where the error passes through regimes that favour different members;
1401 at truncation level a single well-chosen smoother suffices. For the constant-coefficient periodic test
1402 problems the discrete operator is diagonalised by the FFT, and the resulting direct solve, reported in
1403 every table, is faster than every iterative method we measured, including ours ([pending]–[pending]
1404 against the best router); the hybrid is meant for problems on which no such direct diagonalisation
1405 exists—where the band-limited inverse has to be assembled or fitted rather than read off a symbol,
1406 and the corrector is then inexact—which is why Section E also reports a variable-coefficient diffusion
1407 problem.

1408 Additionally, the effectiveness of the learned router depends on the neural operator exhibiting
1409 heterogeneous performance across samples. When the neural operators produces predictions of
1410 relatively uniform quality, the benefit of adaptive routing diminishes and it is more practical to set a
1411 fixed schedule based on the quality of the neural operator. In practice, achieving such heterogeneity
1412 can be sensitive to training choices, and neural operators themselves can be finicky to train.

1413 G LLM Usage

1414 LLMs were used substantially in preparing this paper, and we disclose this precisely. The text of
1415 the original submission was drafted by the authors, with ChatGPT and Gemini used for feedback
1416 on grammar and clarity. The revision that responds to the reviews was produced with Claude Code
1417 (Anthropic), operated and directed by the authors: the wall-clock benchmark pipeline (the compiled
1418 stencil kernels, the cost-aware macro-action environment, the scalar-feature router and its training
1419 loop, the corrector fitting, the baseline, table and figure generators, and the assumption and theorem
1420 checkers) was written largely by the assistant under the authors’ direction and was reviewed, tested
1421 and run by the authors; the corrections to the theory in this revision (the μ -slack in Theorem 4.1, the
1422 corrected constant and hypotheses of Proposition 4.2, the counterexample to postfix monotonicity,
1423 the unitary-eigenbasis condition, and the restatement of Theorem 5.1 for normalised weights) were
1424 drafted with the assistant’s help in response to reviewer findings and were verified by the authors;
1425 and much of the new text of Section 7 and Section E was drafted by the assistant from the generated
1426 tables and edited by the authors. The theorems and proof strategies of the original submission predate
1427 the use of the assistant. The research questions, the framing of the method, and the decisions about
1428 what to run and what to claim are the authors’. Independent reviews of the manuscript and code were
1429 also obtained from LLM-based reviewers; their findings are addressed in this revision.

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1436 with the best static schedule at truncation level; mixed results against the best fixed schedule
1437 below it; multigrid alone faster below truncation level) with counts and ranges computed
1438 from the tables, and the theoretical claims with their hypotheses (unitary eigenbasis for
1439 postfix monotonicity, measured slack otherwise; per-decision consistency for the full-
1440 information router). At the time of this draft several confirmatory cells are still being
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